



US012410048B2

(12) **United States Patent**
Zdeblick et al.

(10) **Patent No.:** **US 12,410,048 B2**

(45) **Date of Patent:** **Sep. 9, 2025**

(54) **INGREDIENT CONTAINER**

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(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 0 days.

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(21) Appl. No.: **18/649,462**

(22) Filed: **Apr. 29, 2024**

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(65) **Prior Publication Data**

US 2024/0343547 A1 Oct. 17, 2024

European Search Report received for European Patent Application
No. 23209898.8, mailed on Apr. 23, 2024, 4 pages.

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Related U.S. Application Data

(62) Division of application No. 17/989,610, filed on Nov.
17, 2022, now Pat. No. 12,084,334.

(51) **Int. Cl.**
B67D 1/00 (2006.01)

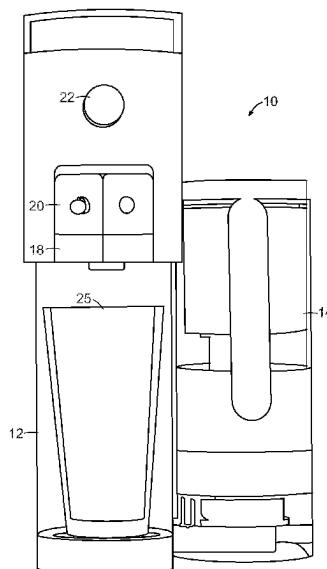
(52) **U.S. Cl.**
CPC **B67D 1/0078** (2013.01)

(58) **Field of Classification Search**
CPC .. B67D 1/0078; B67D 1/1238; B67D 1/0857;
B67D 1/0888; B67D 1/0875; B67D
1/0068; B67D 1/0069
USPC 99/275, 276, 287, 295
See application file for complete search history.

(57) **ABSTRACT**

Ingredient containers for use in a beverage carbonation
system are provided. The ingredient container includes a
container body having a hollow interior and an opening
leading into the hollow interior. The ingredient container
also includes a cap positioned over the opening in the
container body and including an inlet port and an outlet port.
A cross-section of the cap extends substantially perpendicu-
lar to a central axis of each of the inlet port and the outlet
port, and the cross-section of the cap has a shape that is a
substantially circular triangle.

15 Claims, 26 Drawing Sheets



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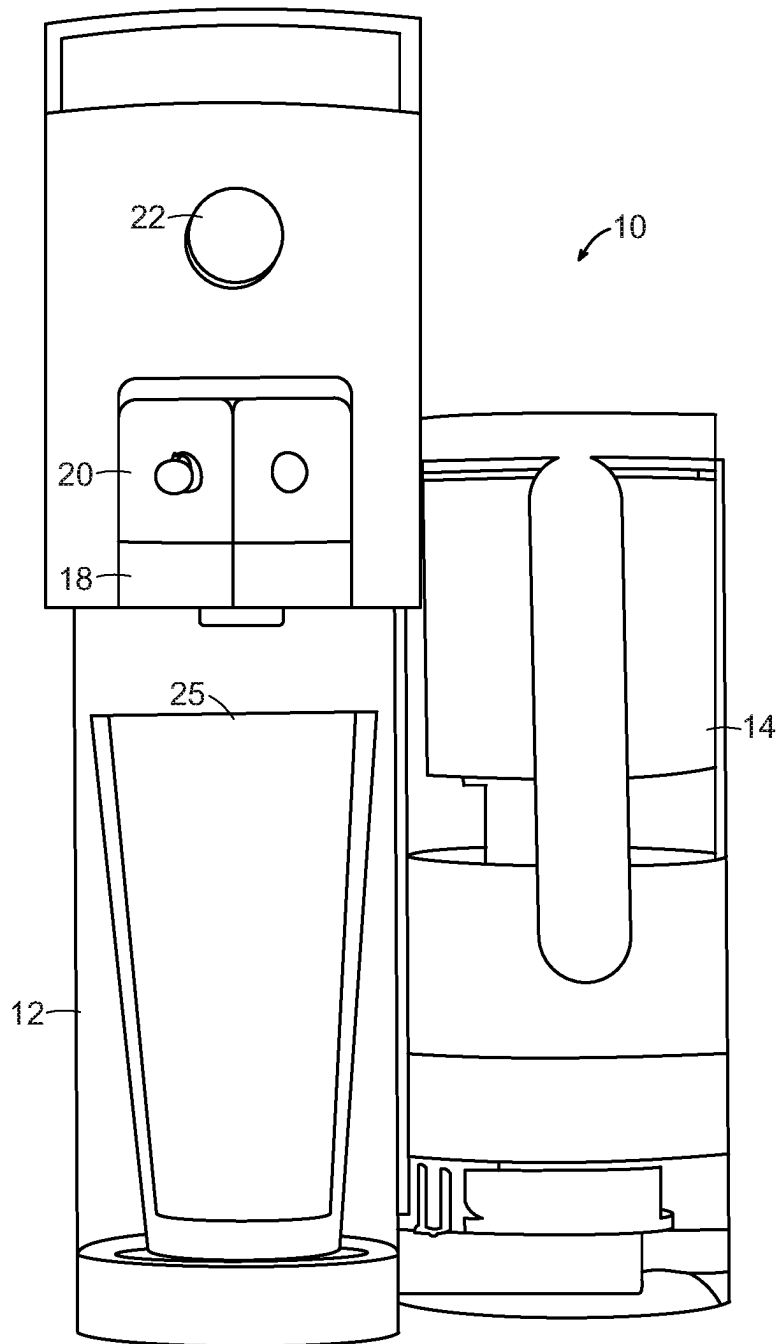


FIG. 1

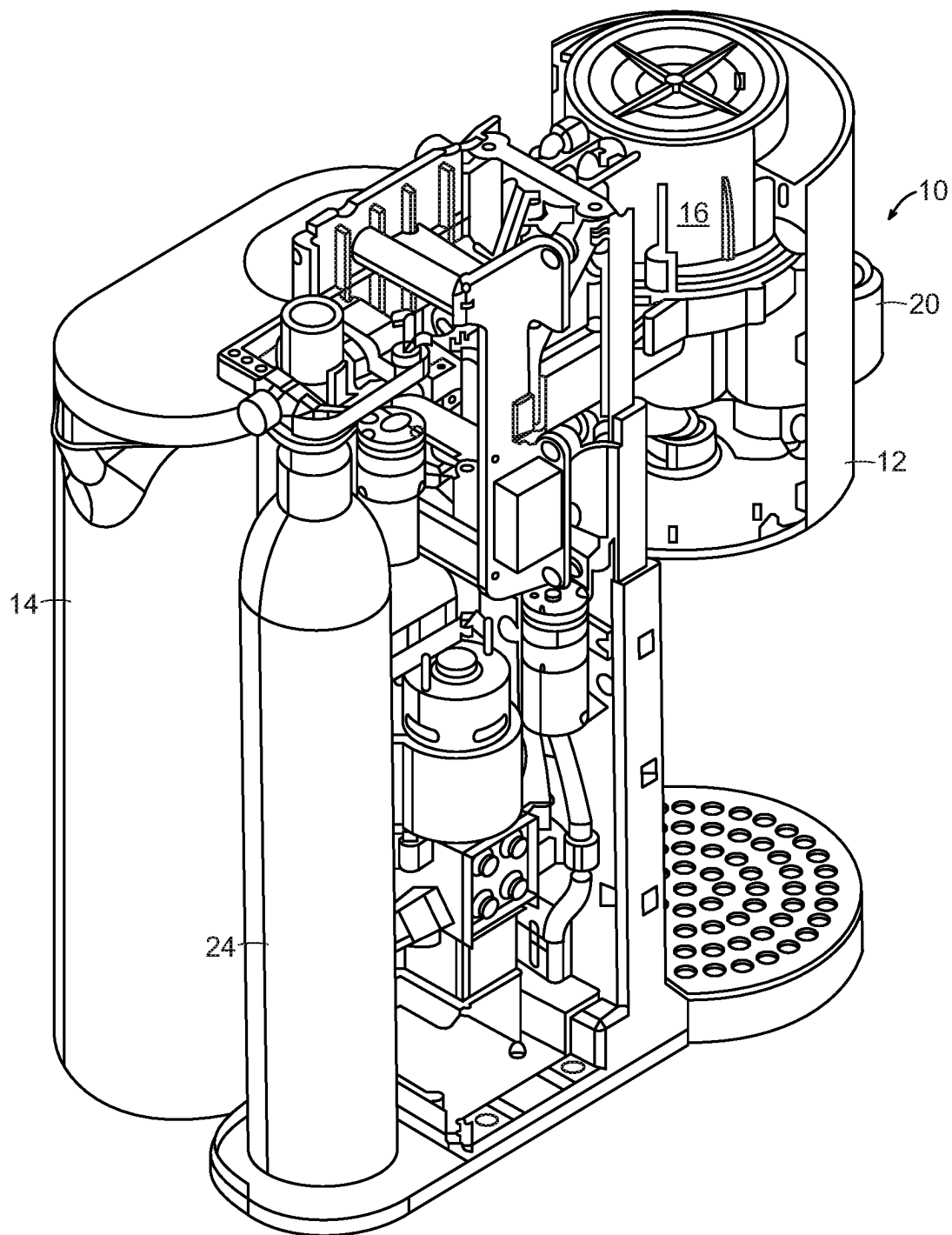


FIG. 2

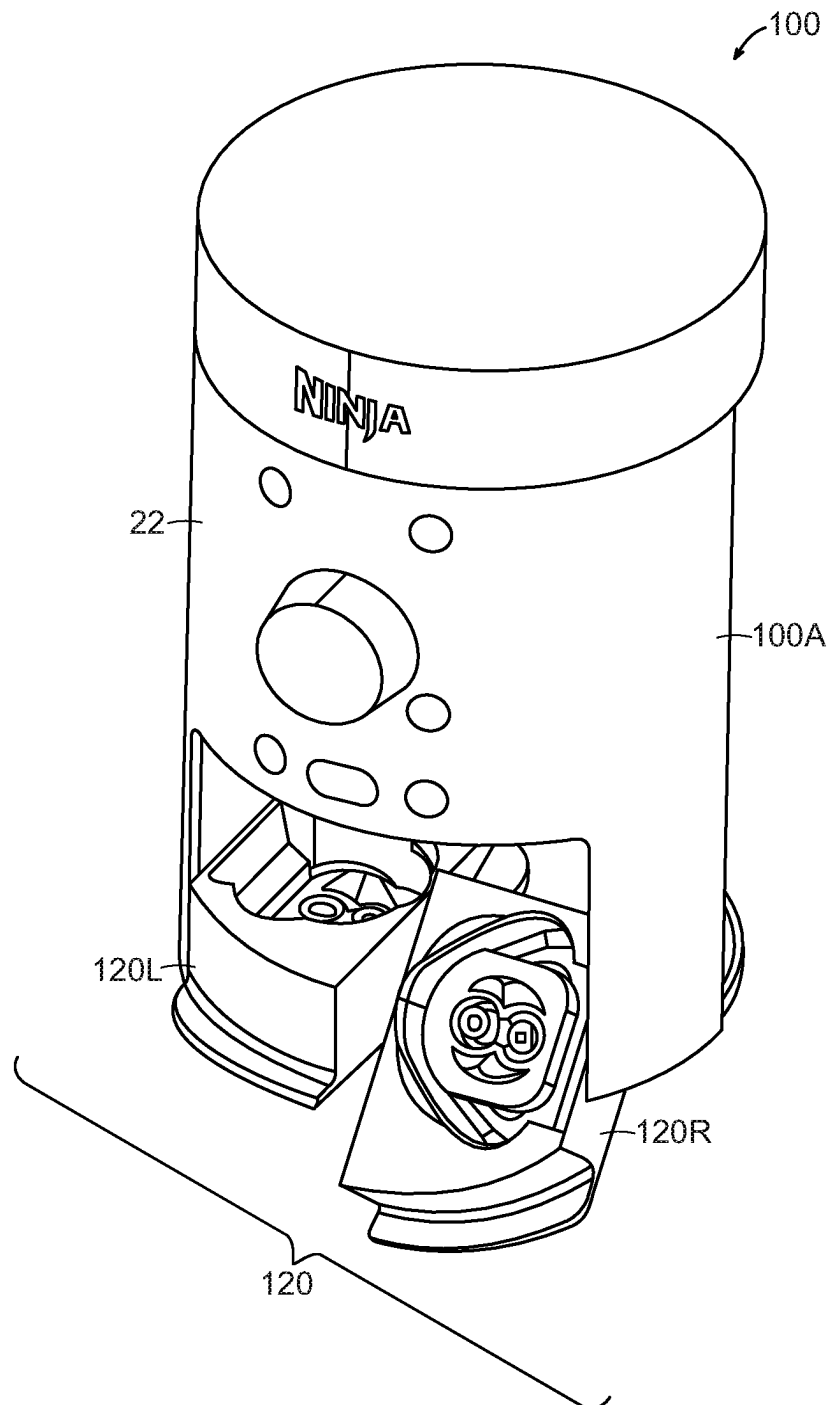


FIG. 3

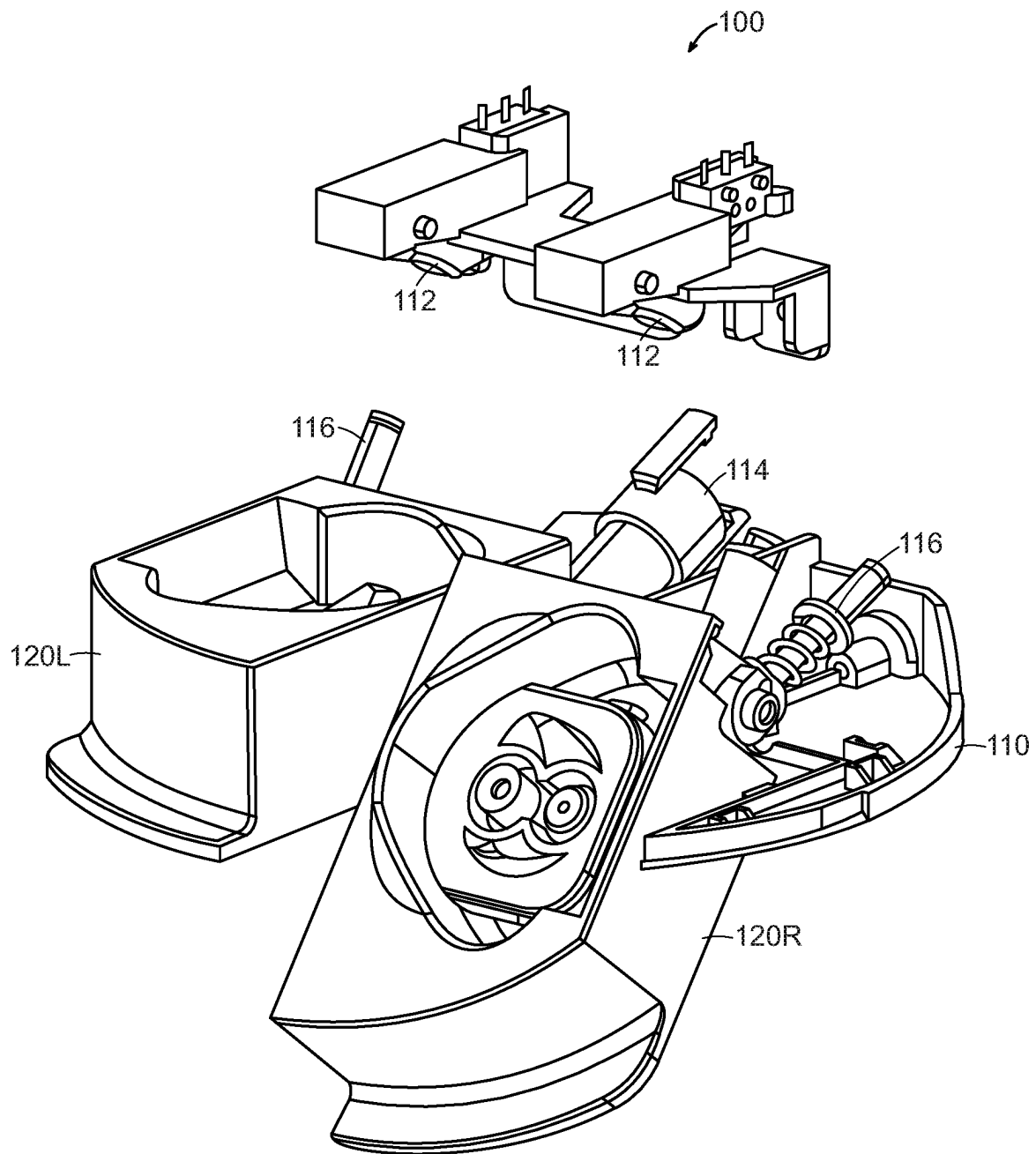


FIG. 4

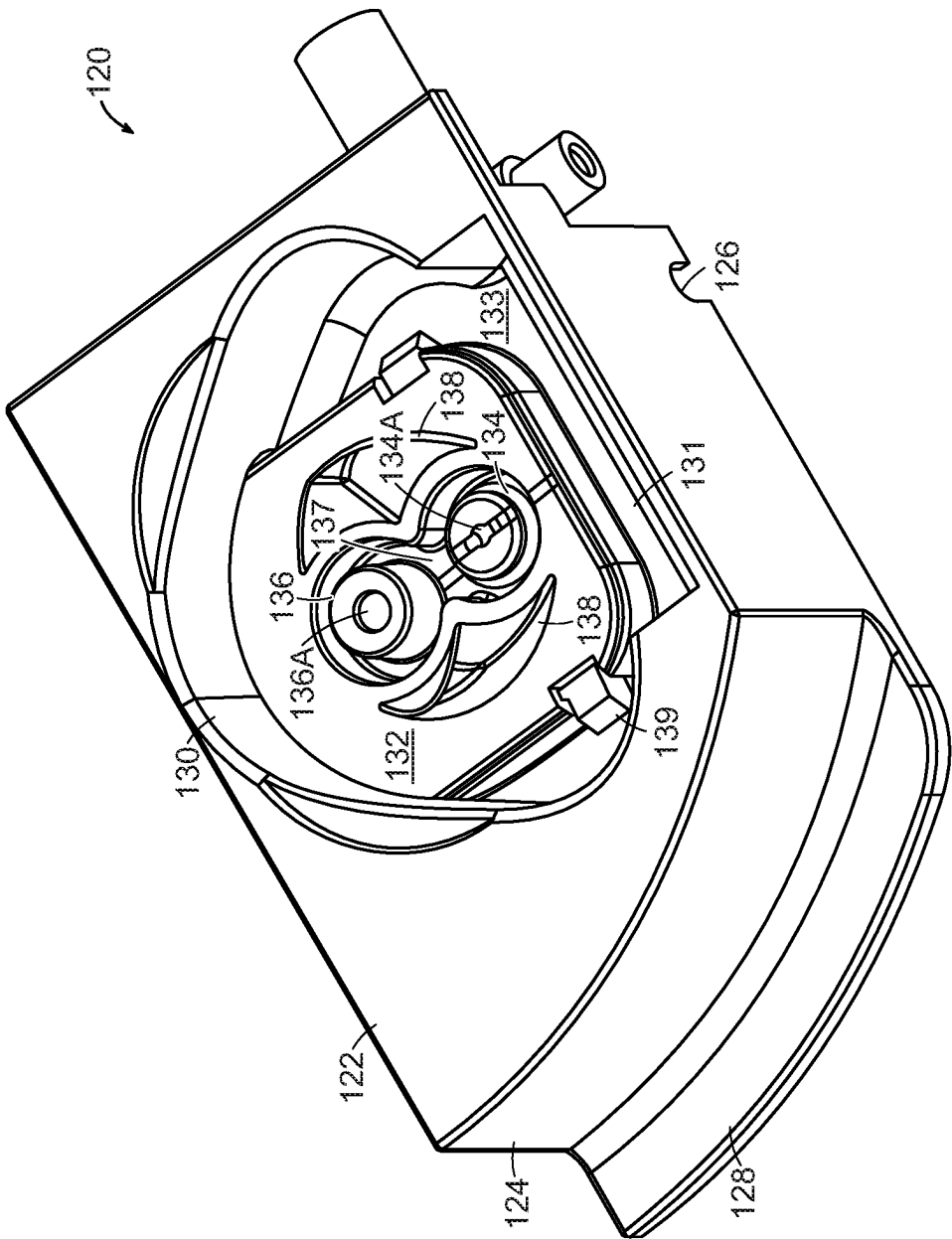


FIG. 5A

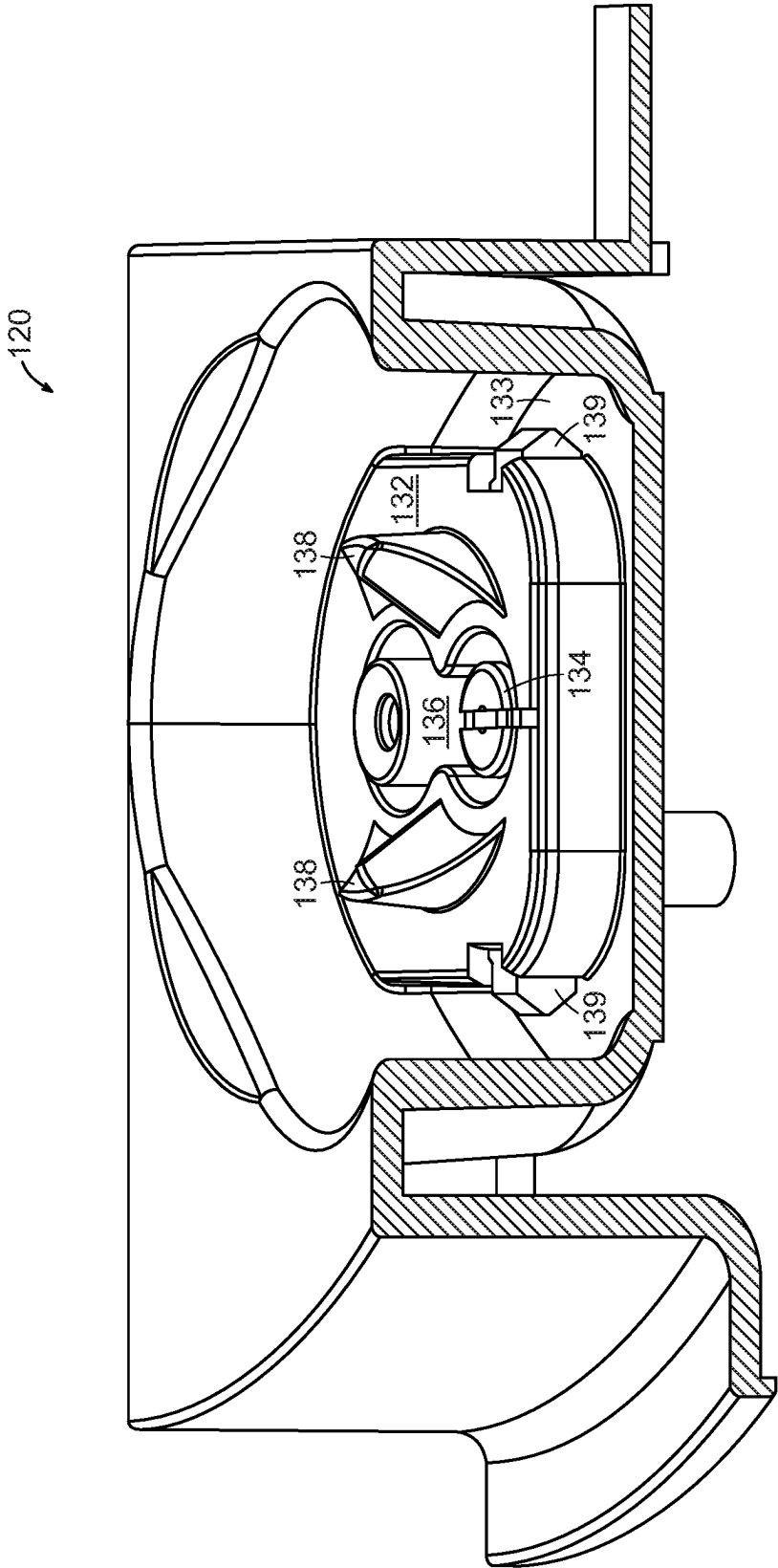


FIG. 5B

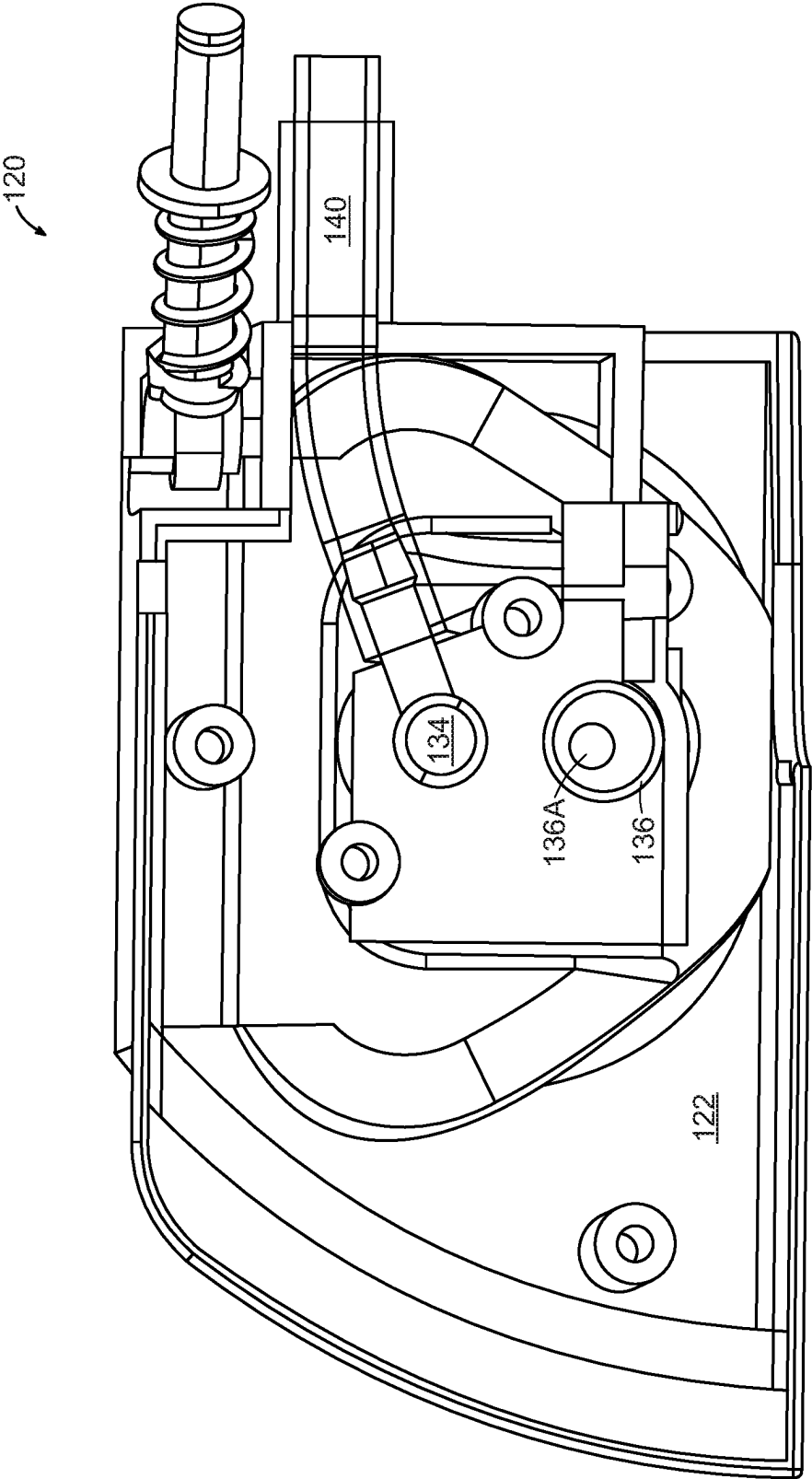


FIG. 6

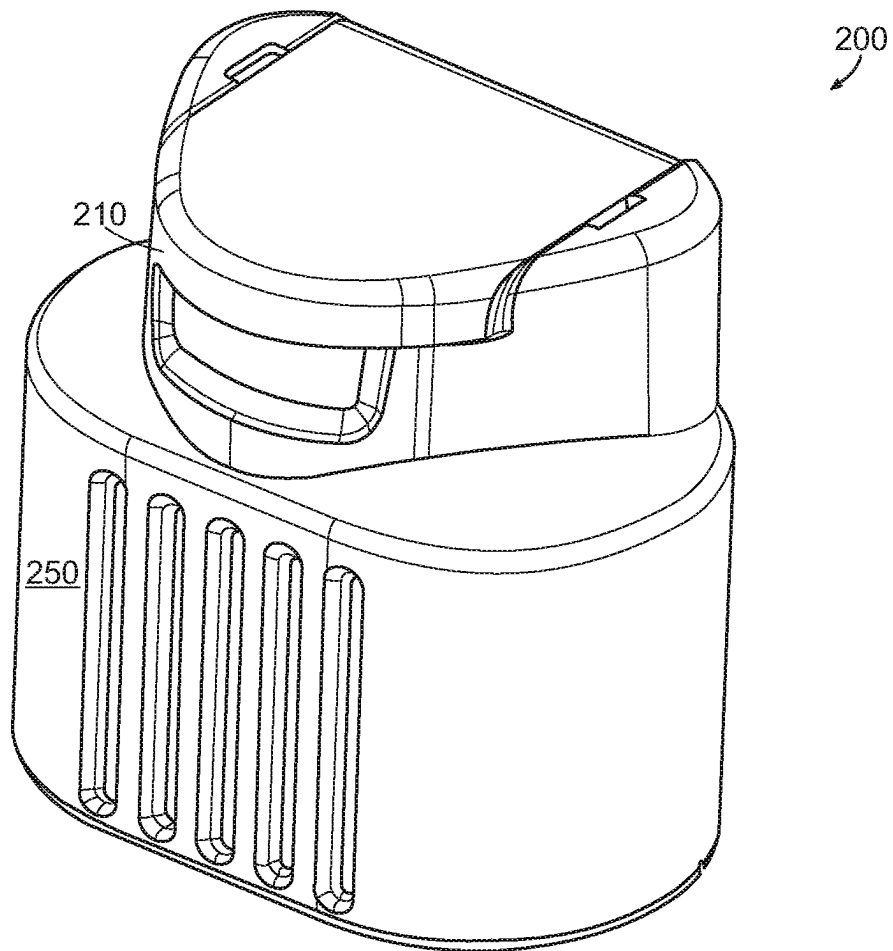


FIG. 7

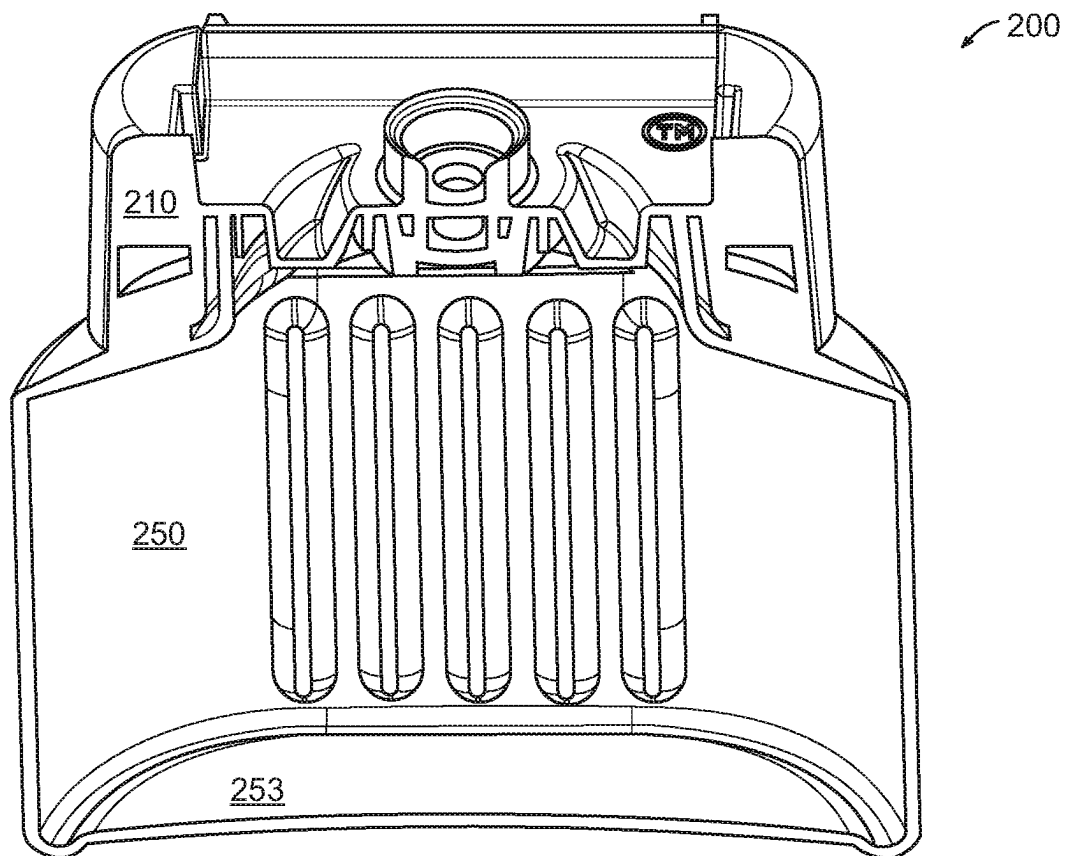
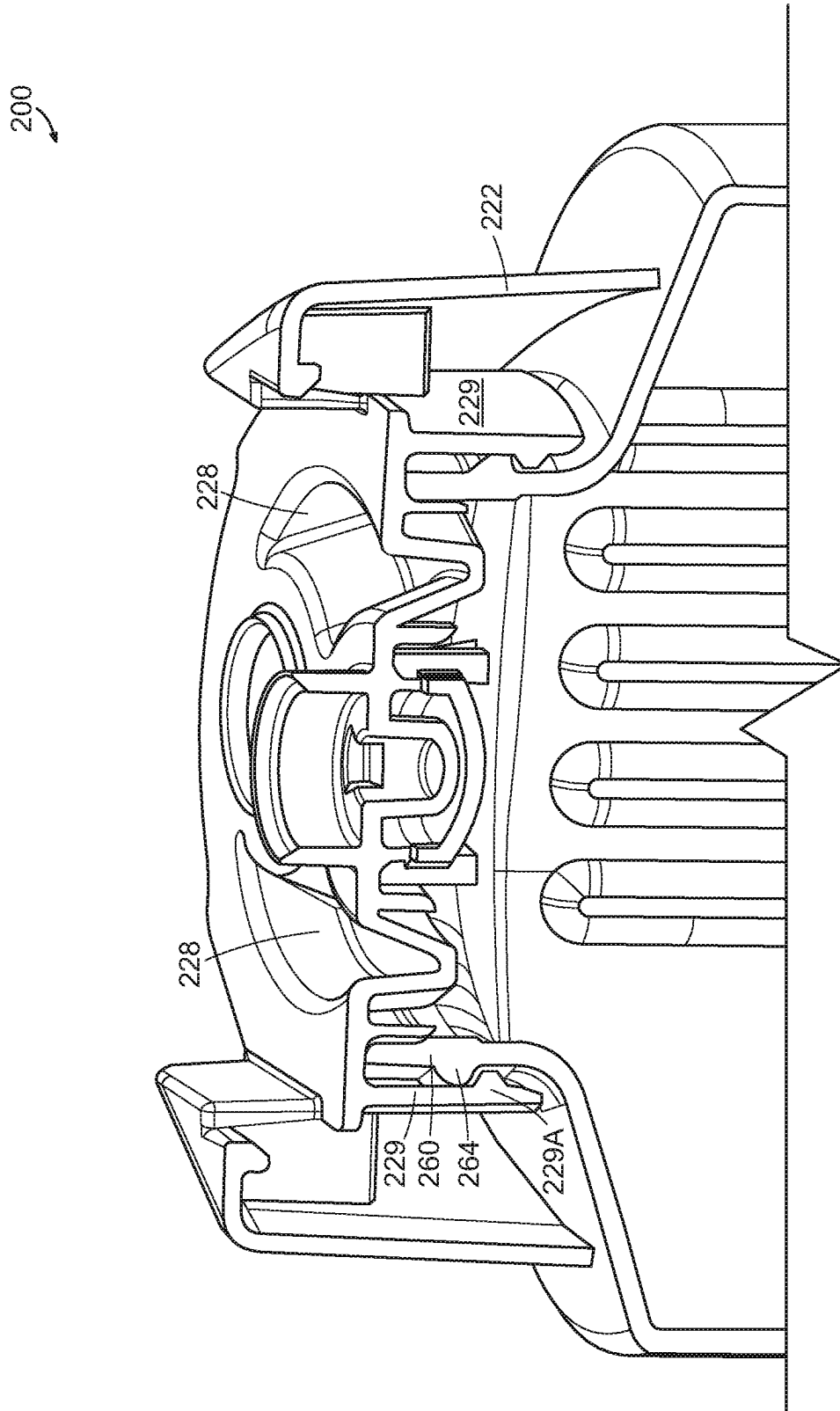


FIG. 8A



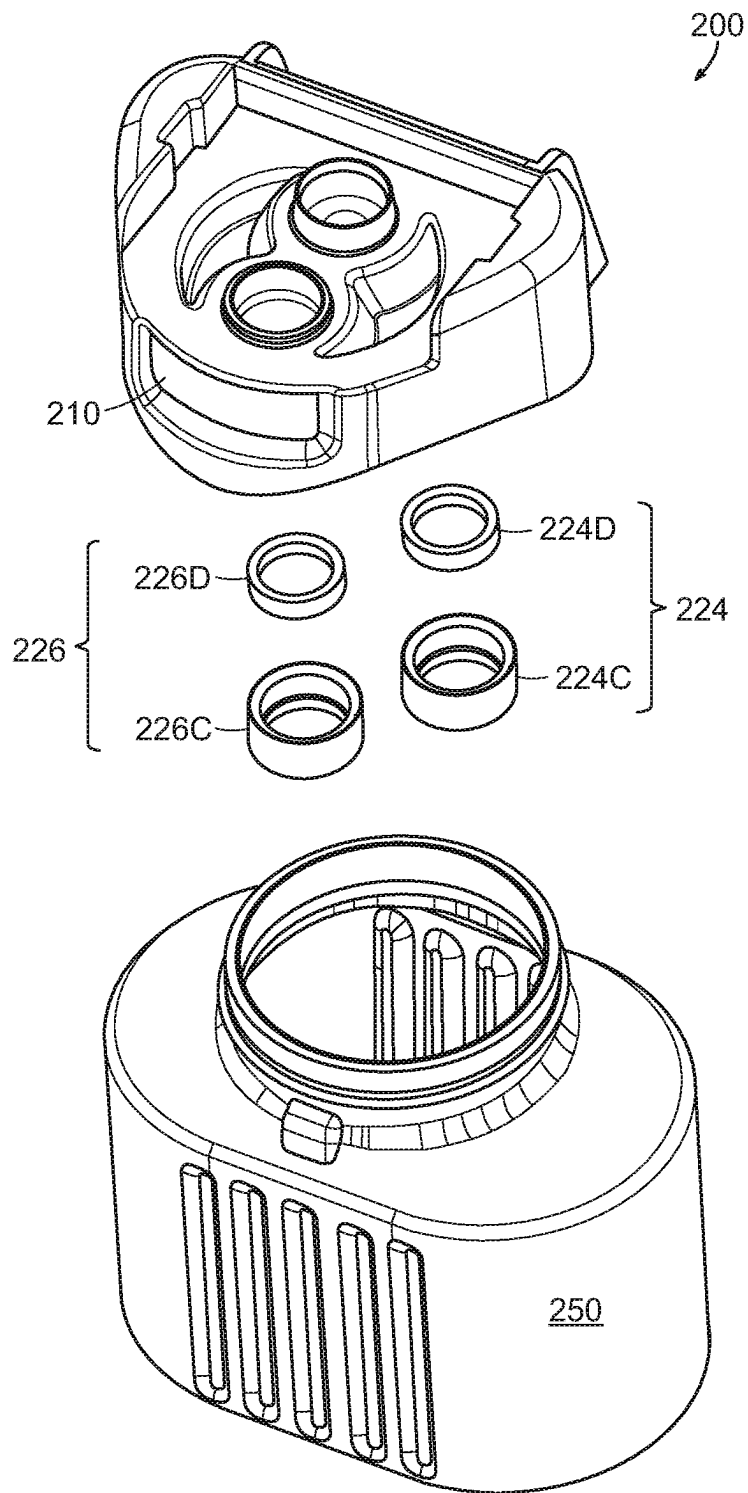


FIG. 9

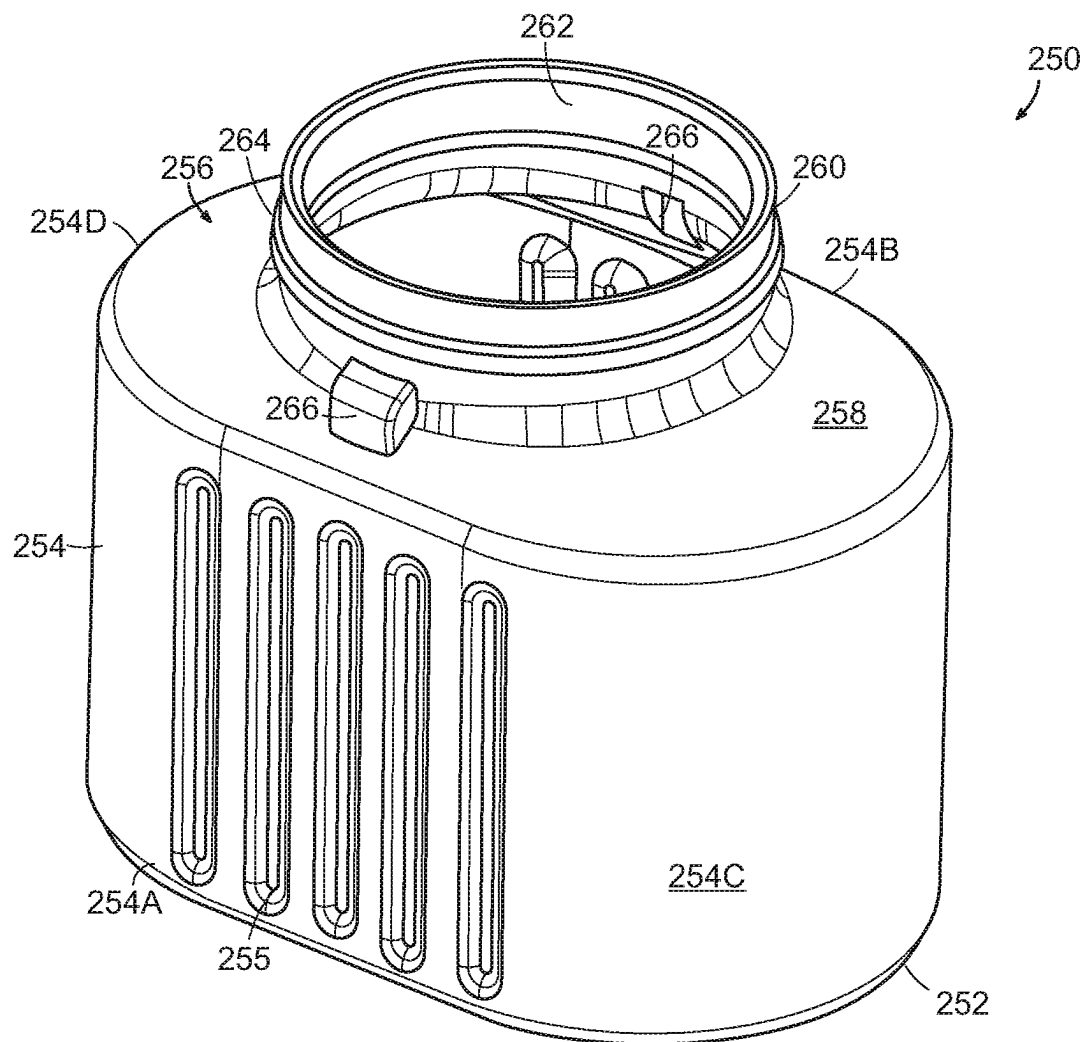


FIG. 10A

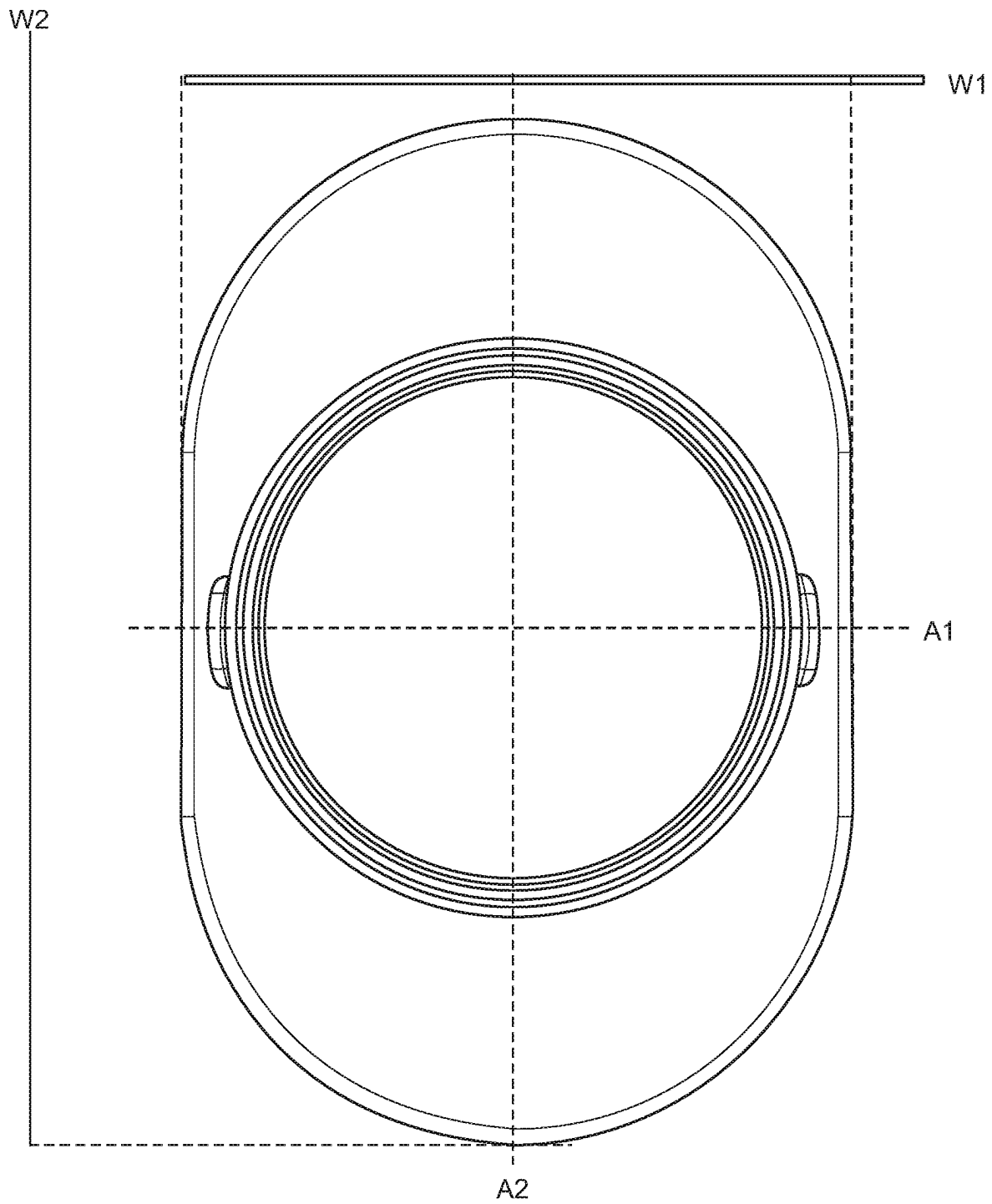


FIG. 10B

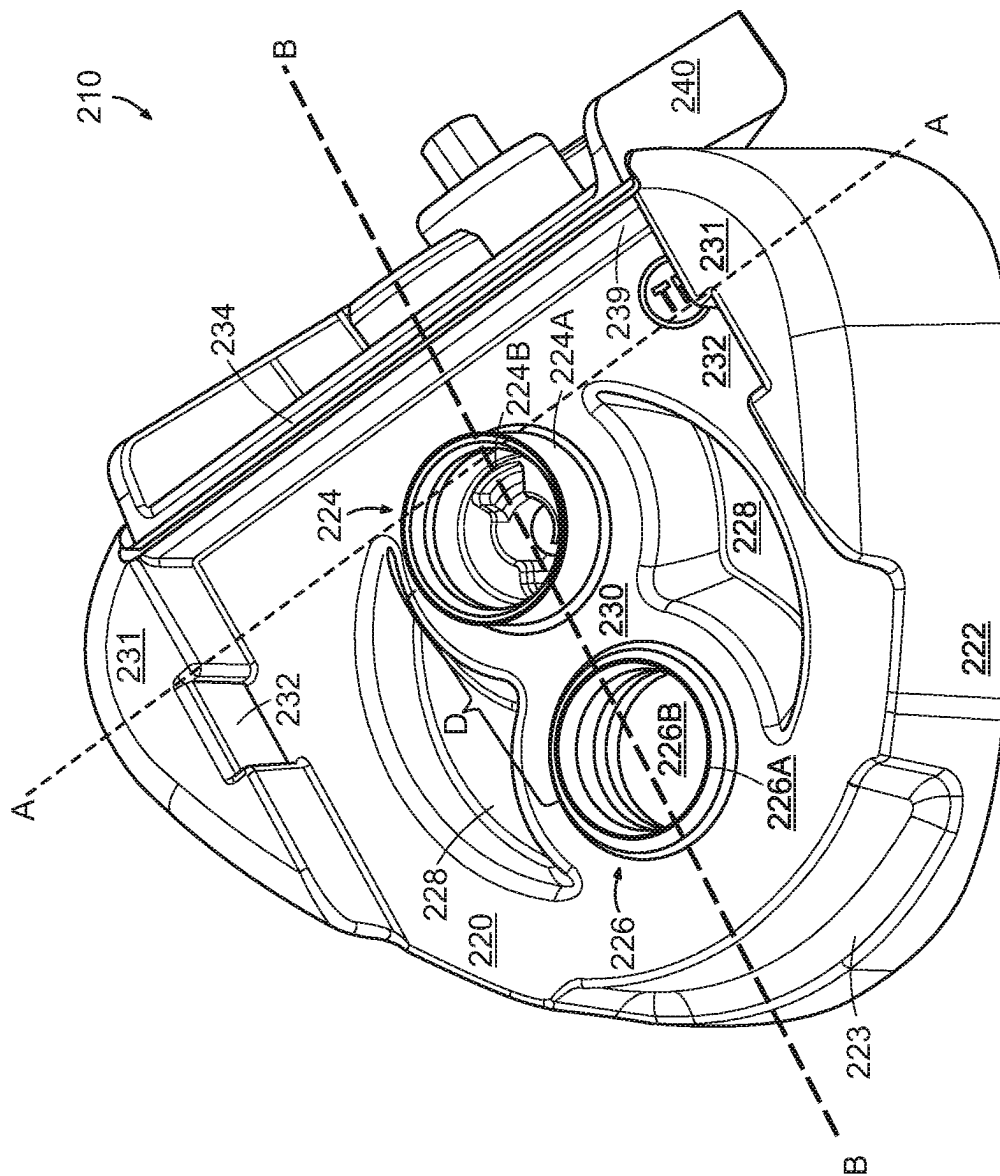


FIG. 1A

FIG. 11B

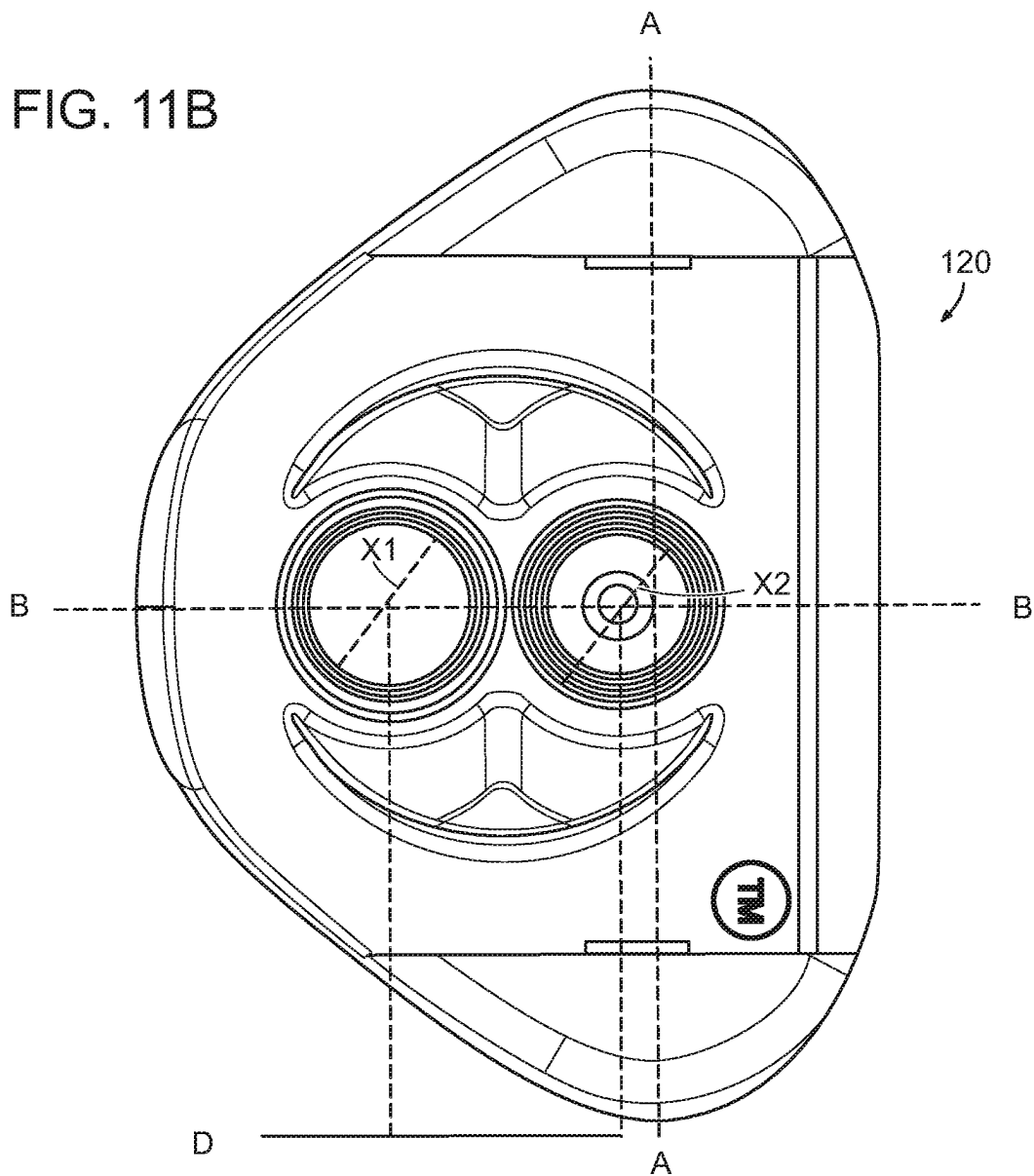
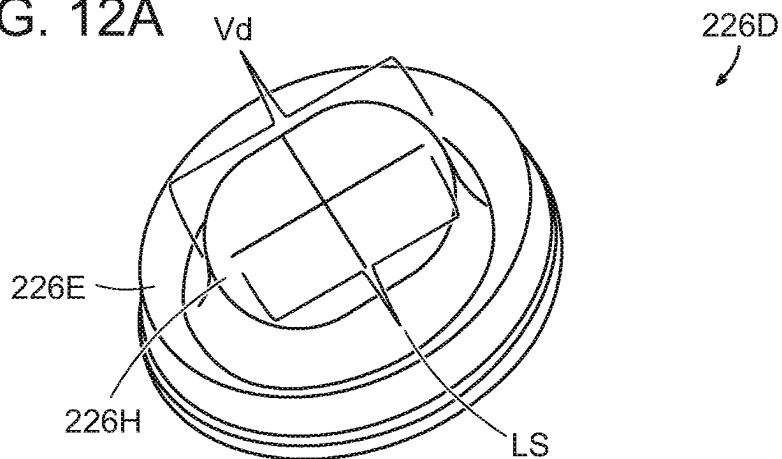


FIG. 12A



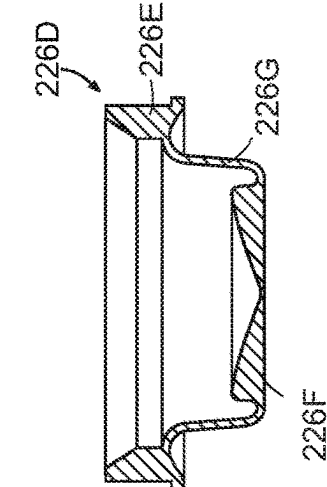


FIG. 12B

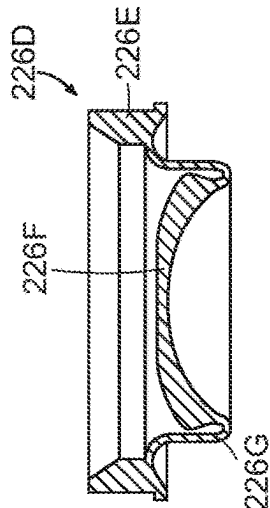


FIG. 12C

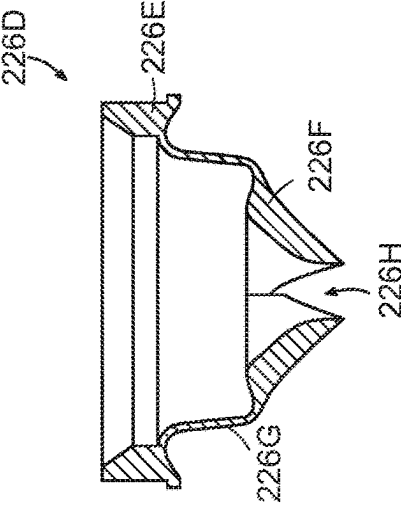


FIG. 12E

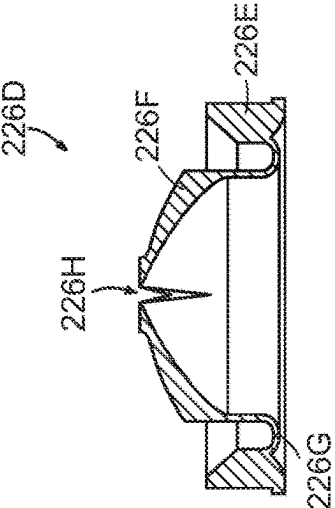


FIG. 12F

FIG. 12D

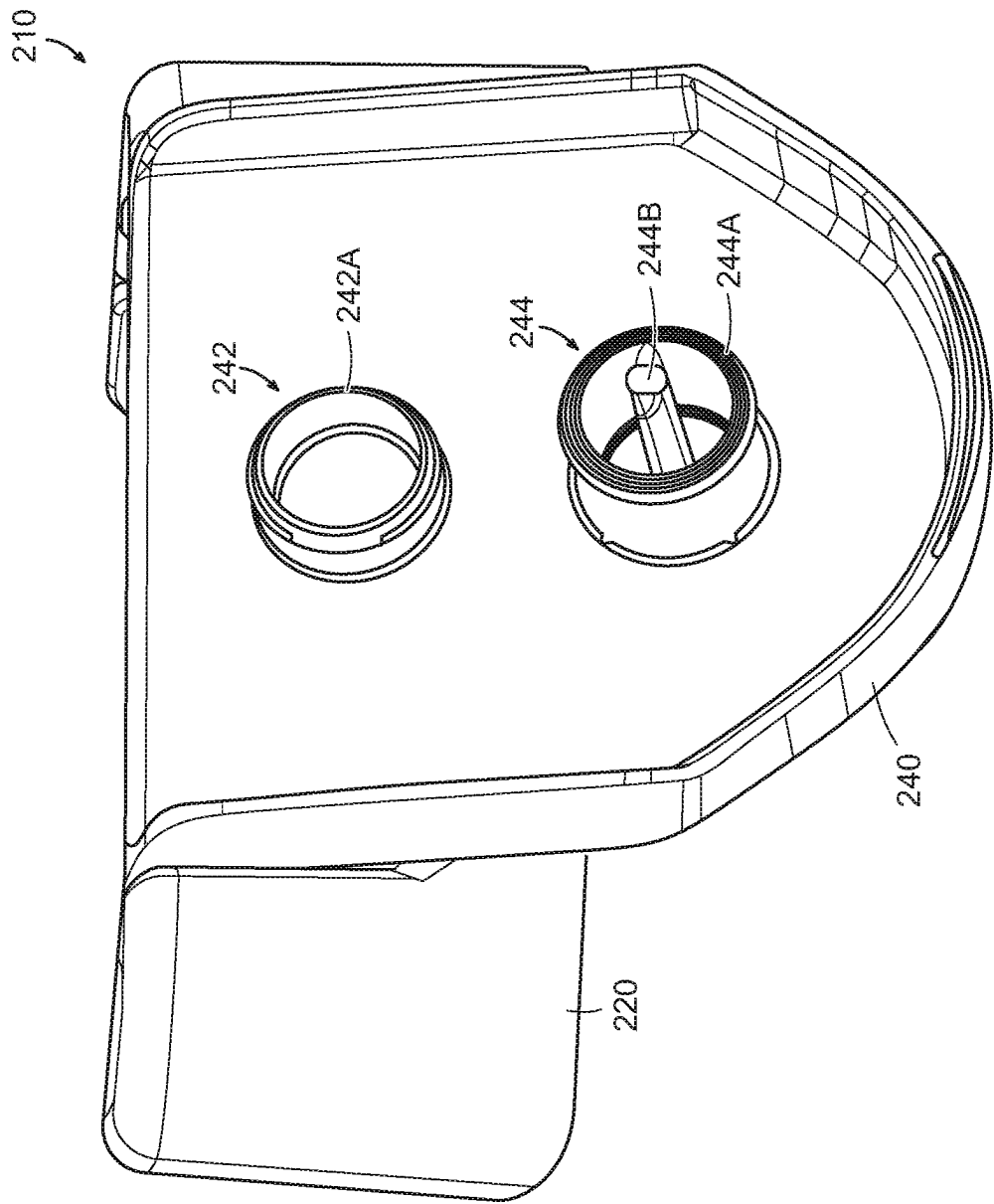


FIG. 13

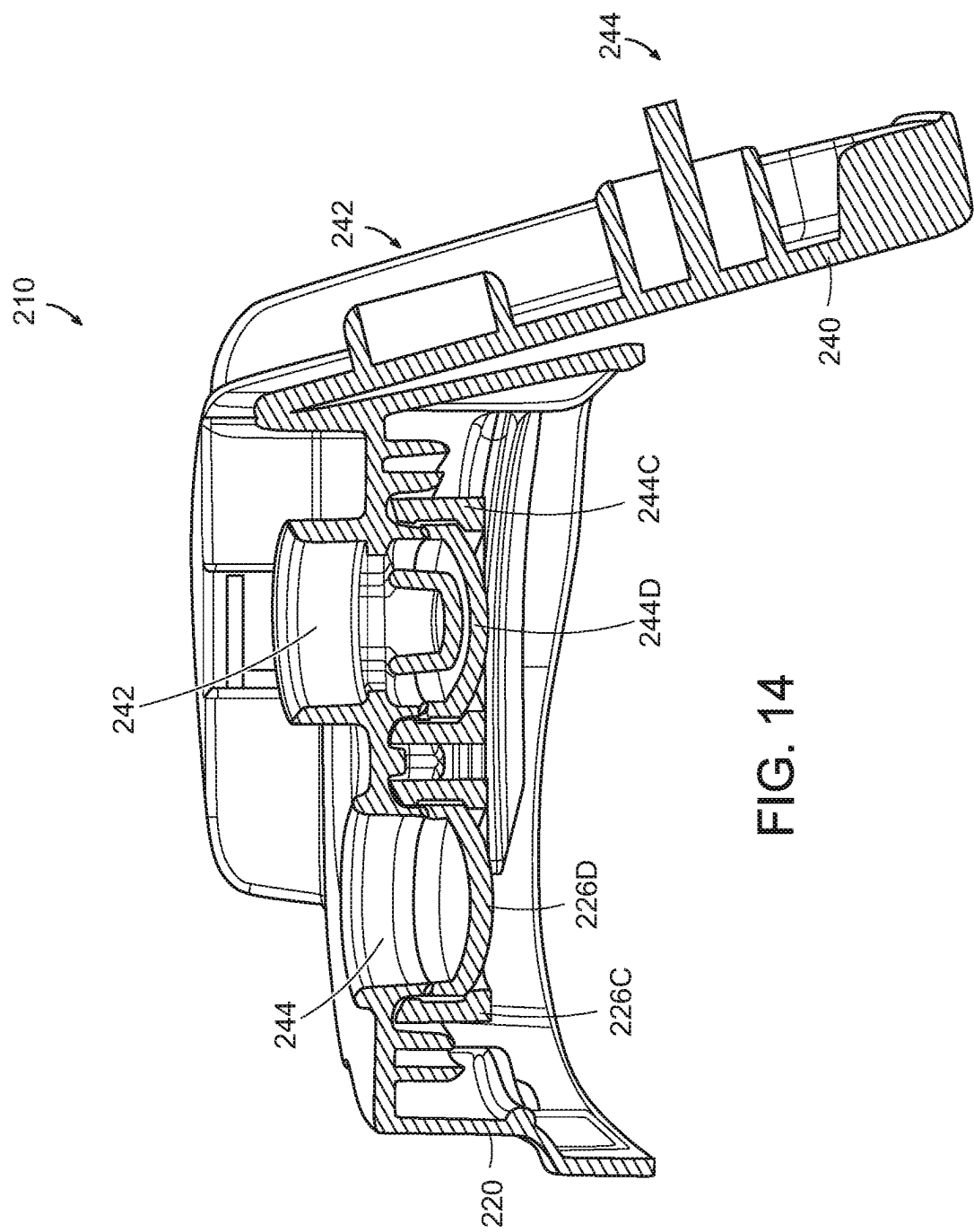


FIG. 14

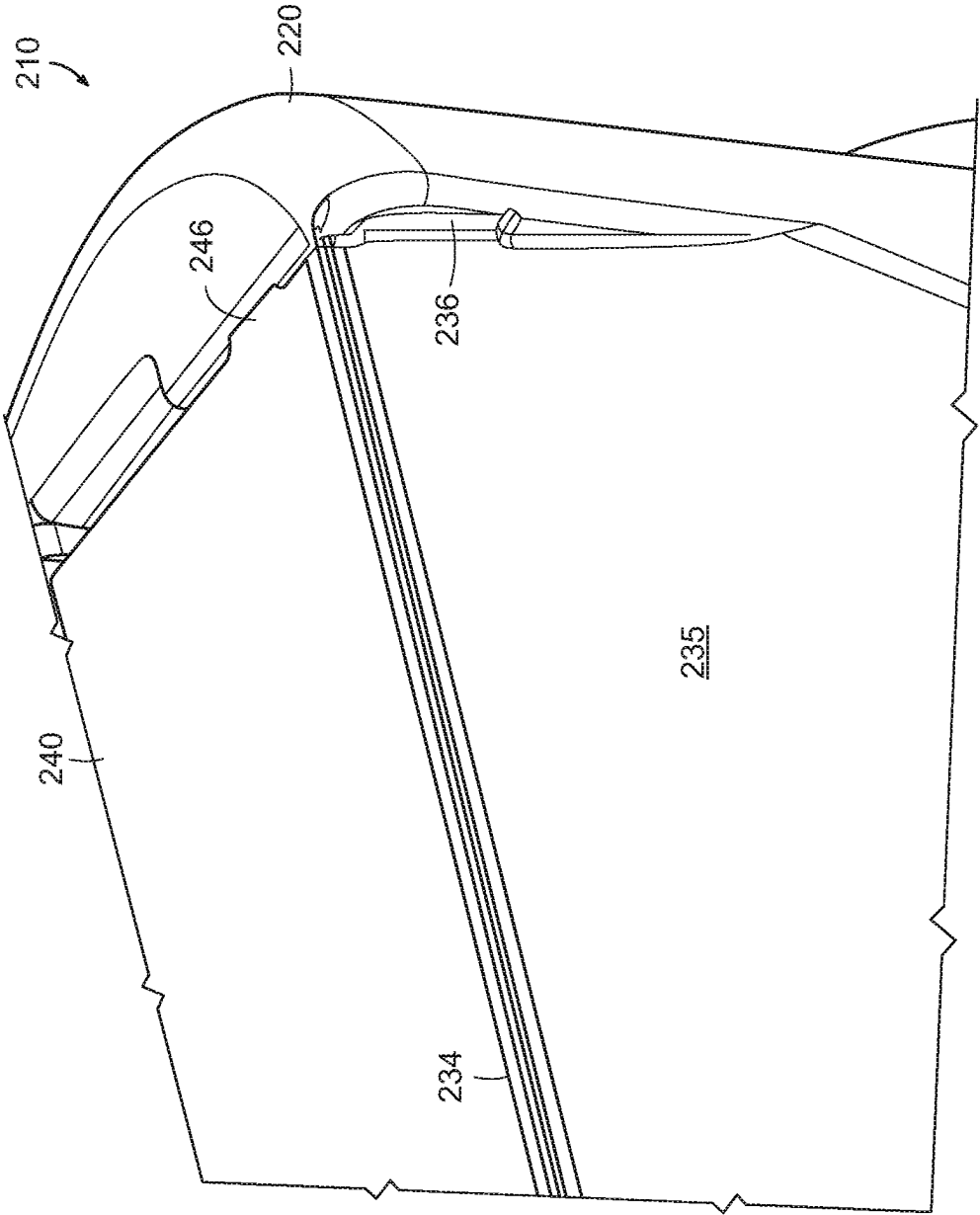


FIG. 15

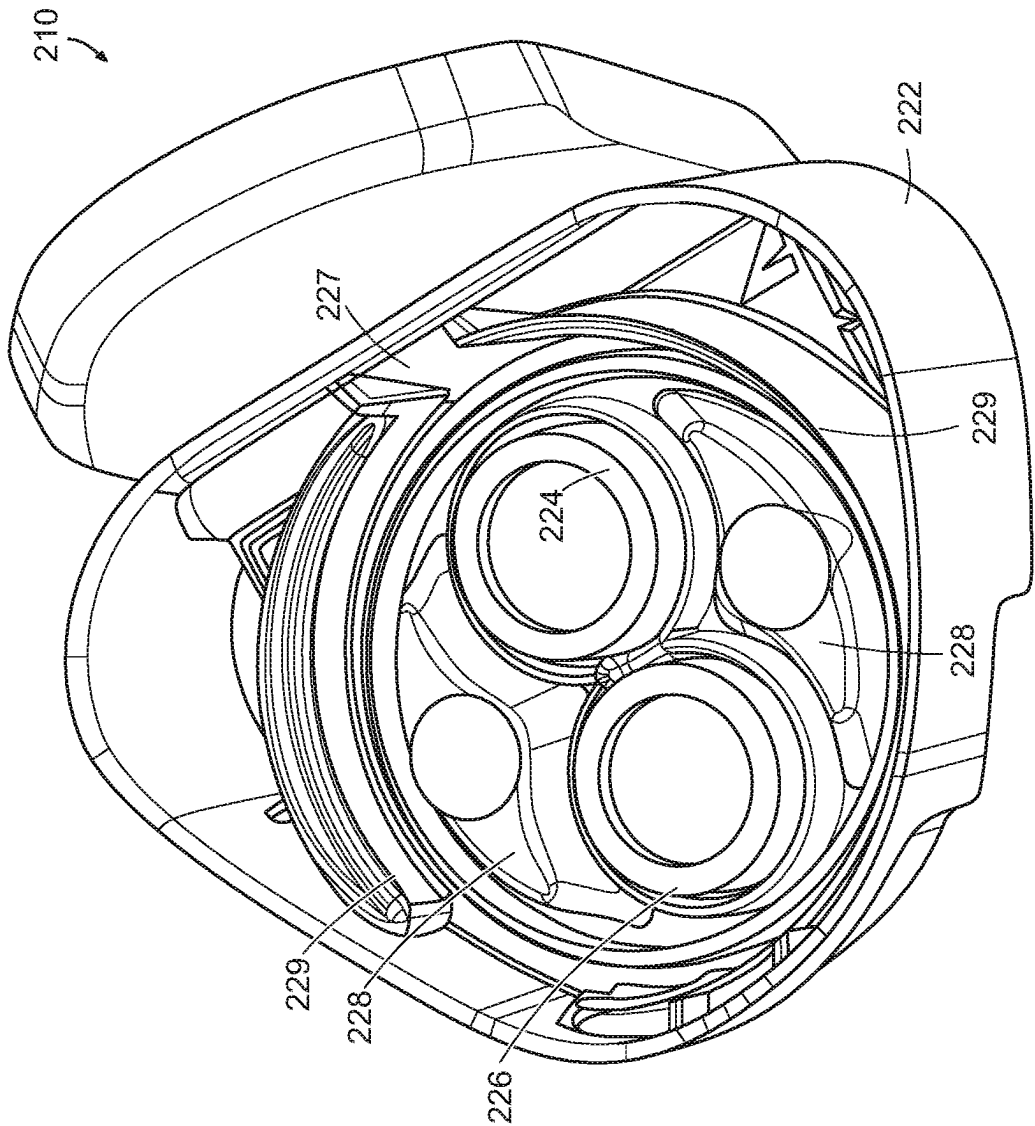


FIG. 16

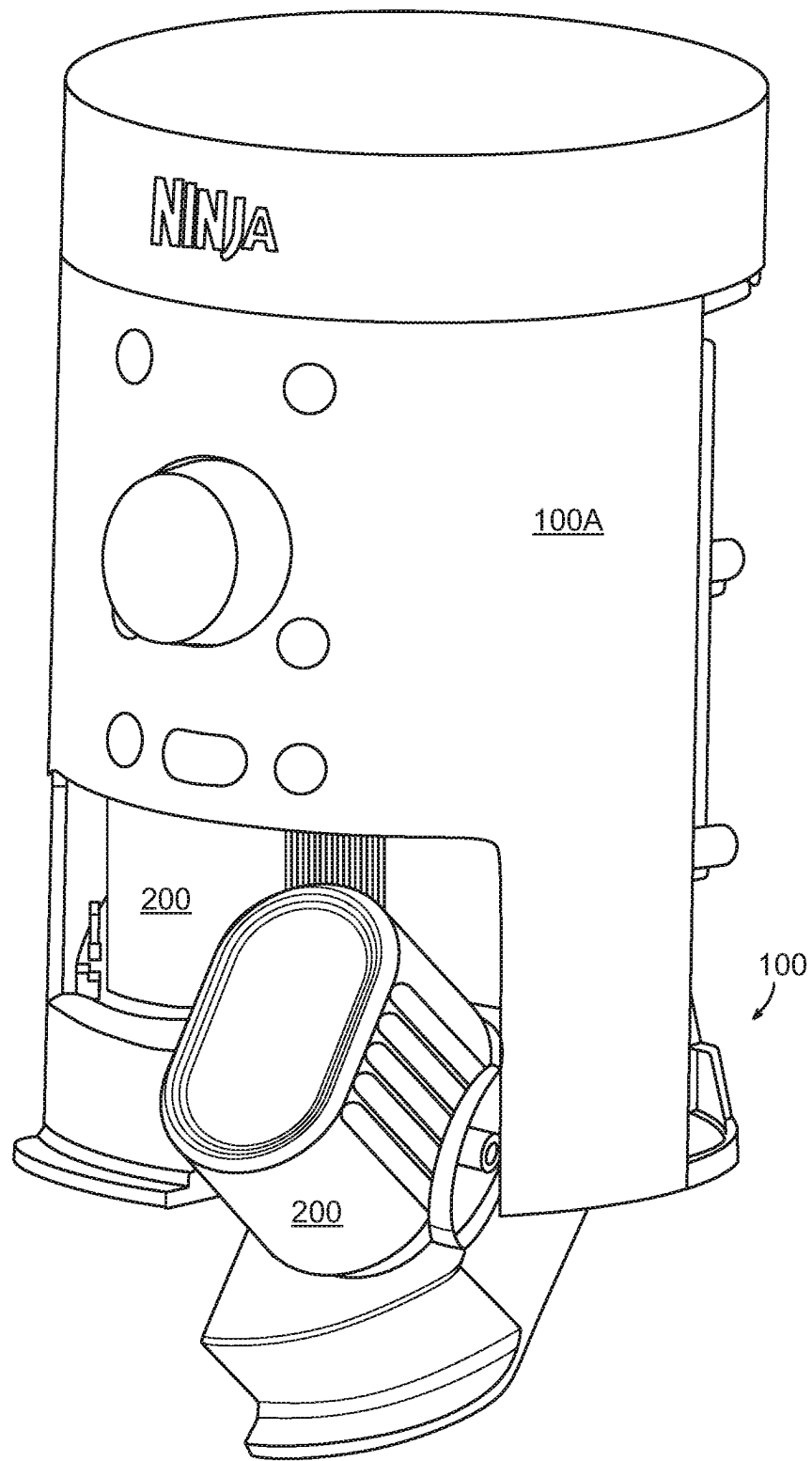


FIG. 17

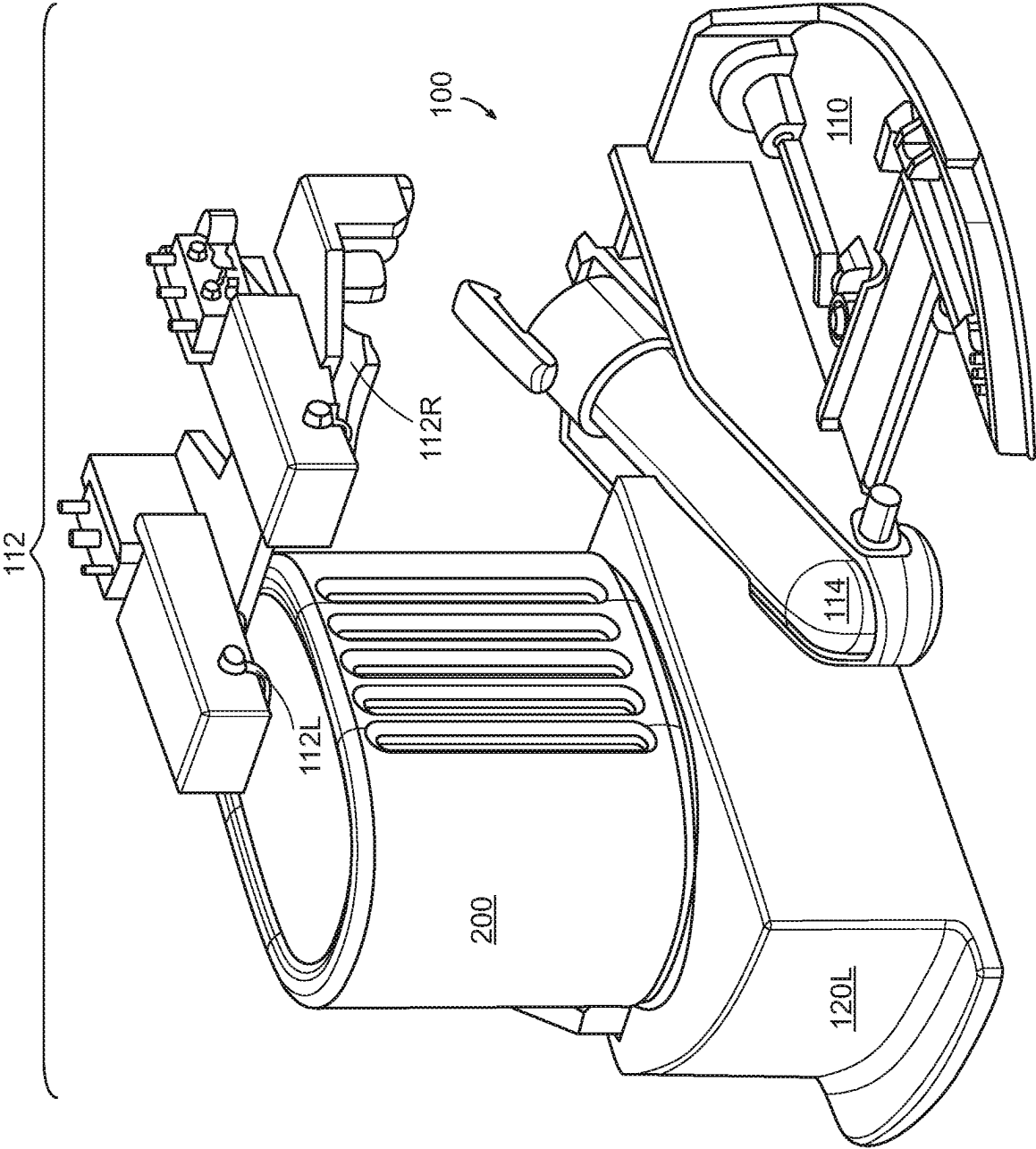


FIG. 18

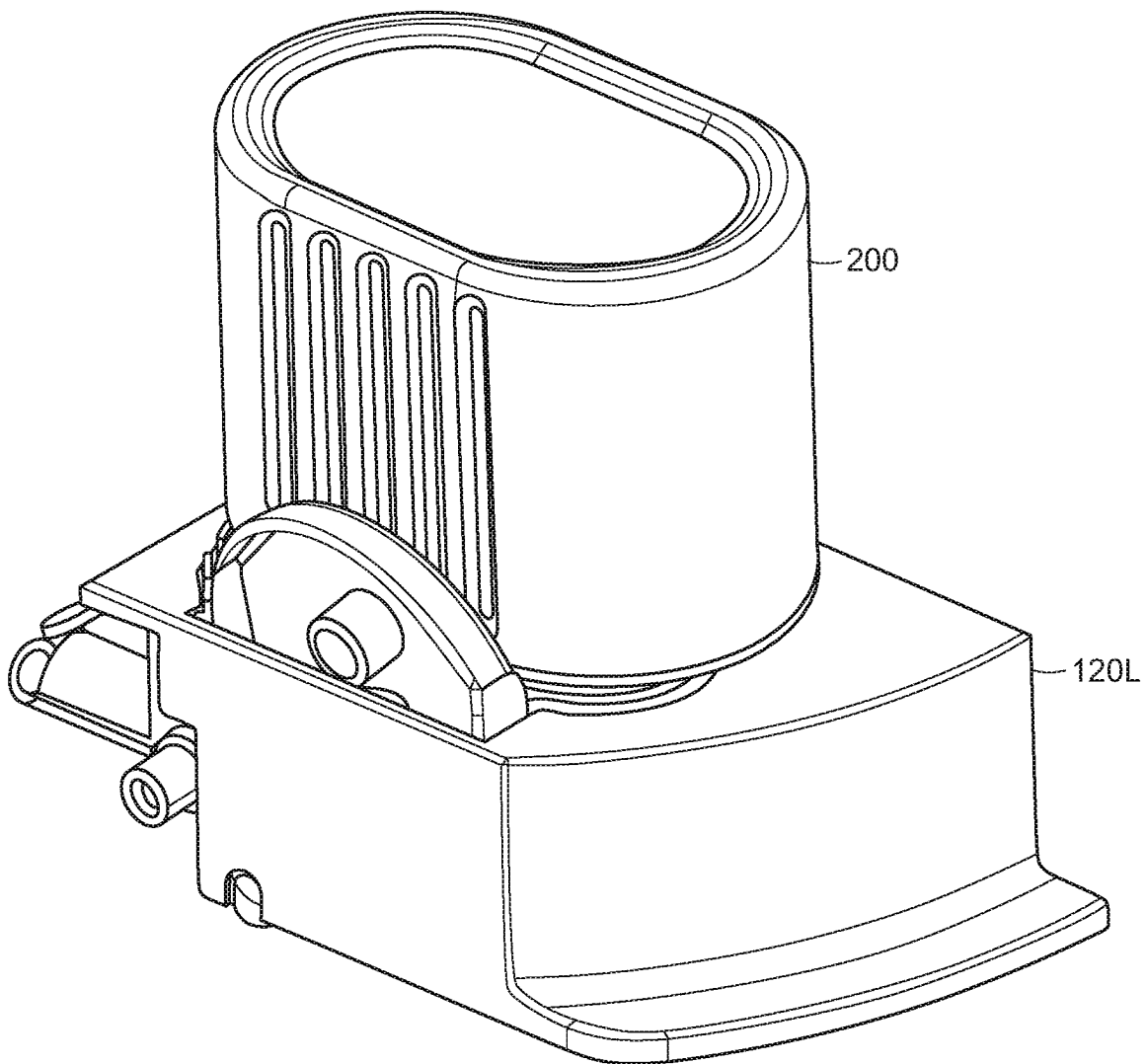


FIG. 19

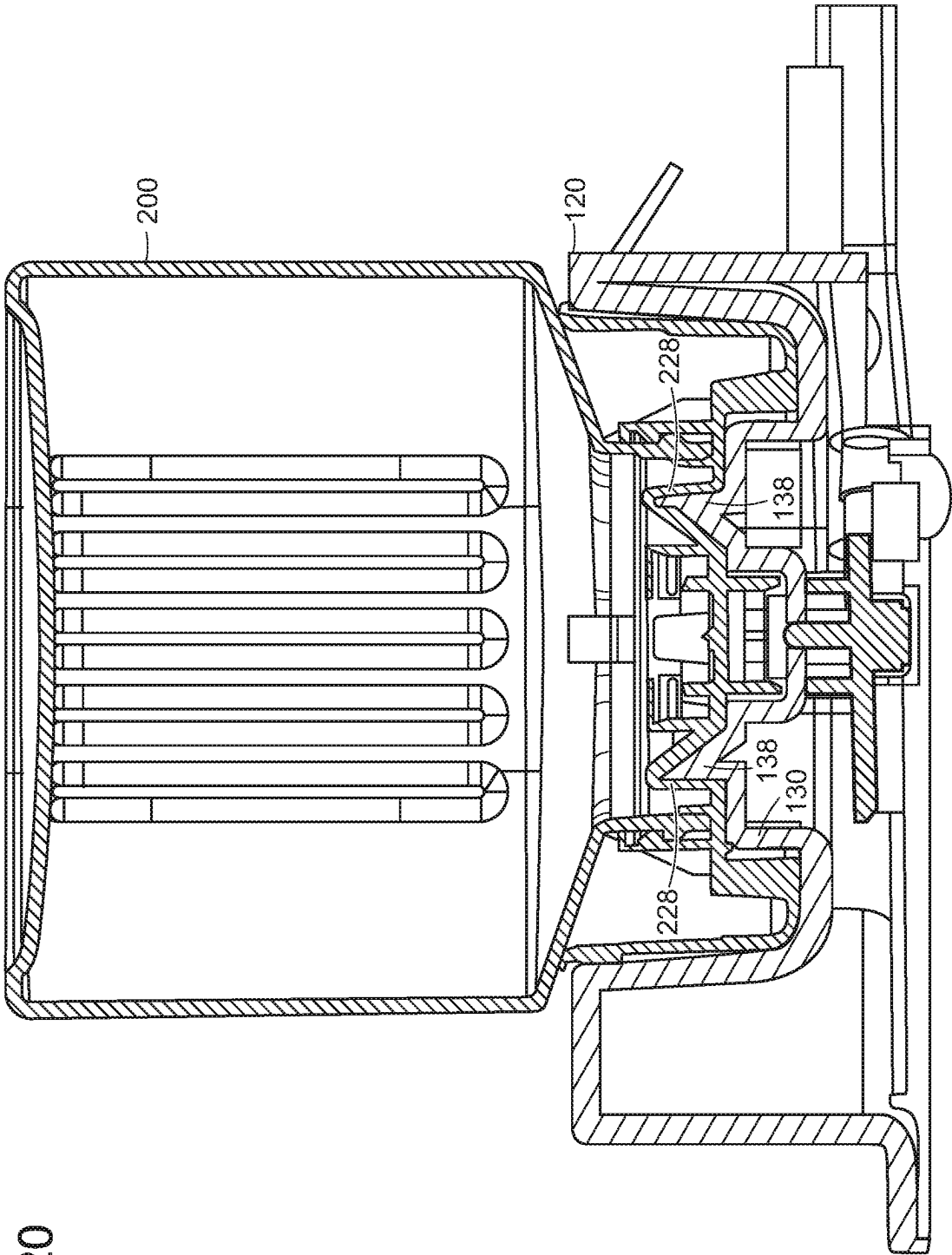
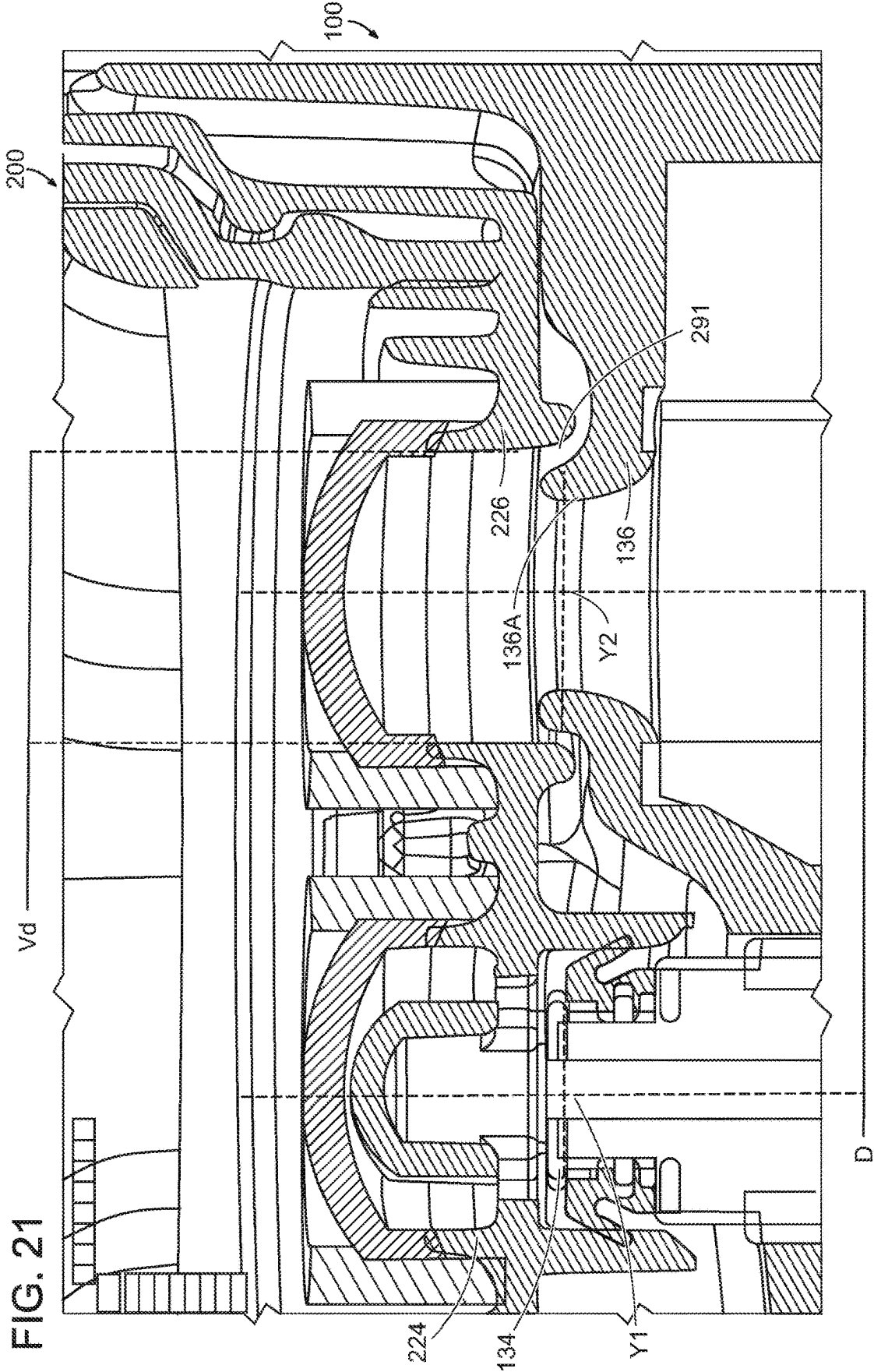


FIG. 20



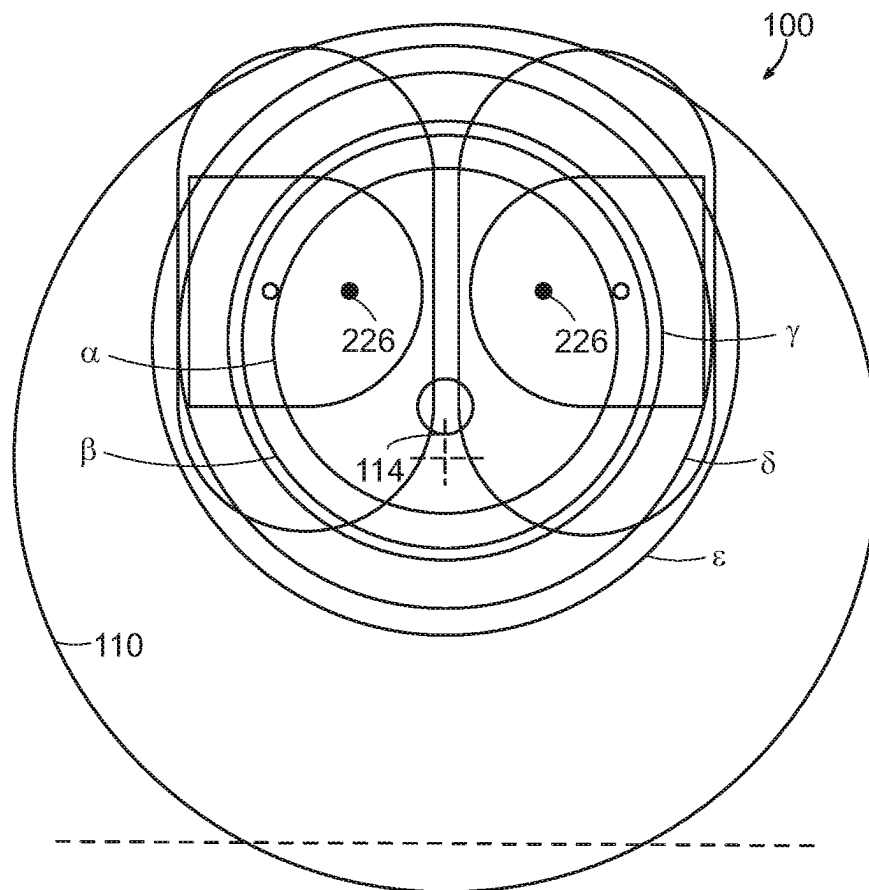


FIG. 22

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INGREDIENT CONTAINER**CROSS-REFERENCE TO RELATED APPLICATIONS**

This application is a divisional of U.S. patent application Ser. No. 17/989,610, filed on Nov. 17, 2022, entitled "Ingredient Container," which is hereby incorporated by reference herein in its entirety.

FIELD

Ingredient containers used with beverage dispensing devices are provided.

BACKGROUND

Conventional beverage dispensing devices operate to carbonate and/or flavor water. Some devices may mix carbonated water and a flavoring compound together in a machine and then dispense the resulting mixture into a receptacle. Unless the devices are thoroughly cleaned, this method can result in contamination occurring over time. Other devices rely on crushing, puncturing, and/or generally compromising flavoring containers in order to access the flavoring compounds inside. These methods of breaching flavoring containers can result in splatter and mess, which, if not thoroughly cleaned, can result in similar contamination.

Still other devices rely on carbonating water within a specialized container to be attached to the device, and from which the resulting beverage is served. The container can be pre-filled with water and/or flavoring, and then it can be secured to the devices and pressurized within the container and used to serve the resulting beverage. These devices, however, can create excess plastic waste, as specially adapted bottles must be produced to interface with the device.

Accordingly, there remains a need for a better beverage dispensing device to improve on mess creation and waste production.

SUMMARY

Ingredient containers for use with beverage dispensing systems are provided. Related apparatuses and techniques are also provided.

In one embodiment, a container is provided and can include a container body defining a hollow interior, and a cap having an end wall with a first collar projecting therefrom and a second collar projecting therefrom. The first collar can have an inlet valve therein, and the second collar can have an outlet valve therein. The first and second collars can be spaced apart from one another. The end wall can further have first and second recesses surrounding the first and second collars. The first and second recesses can be formed in a surface of the end wall.

The container can vary in a number of ways and may include any of the following features, alone or in combination. For example, the first and second collars and the first and second recesses together can define a figure-8 shaped feature. The container can also include first and second shoulder portions positioned on opposite sides of the end wall and projecting outward from the outward facing surface of the end wall. For example, each of the first and second recesses can have first and second curved sidewalls that extend partially around the first and second collars, respectively. In some aspects, each of the first and second recesses

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can have a third curved sidewall positioned opposite the first and second curved sidewalls. For example, the first and second recesses can be positioned on opposite sides of the first and second collars. For example, the cap can have a minor axis and a major axis, and wherein the cap is substantially symmetrical about the minor axis. In some aspects, the first and second collars can be aligned along the minor axis.

In another embodiment, a container is provided and includes a container body defining a hollow interior, and a cap coupled to the container body to close off the hollow interior. The cap can include at least one recess having a figure-8 shaped projection with first and second openings therein. The first opening can include an inlet valve and the second opening can include an outlet valve, and the figure-8 shaped projection can be at least partially defined by first and second recesses formed in a surface of the cap.

The container can vary in a number of ways and may include any of the following features, alone or in combination. For example, the substantially figure-8 shaped projection can include first and second collars defining the first and second openings and that are spaced a distance apart from one another, having the inlet and outlet valves disposed therein. For example, the first and second recesses surrounding the figure-8 shaped projection can each include first, second, and third sidewalls. The first and second sidewalls can be substantially convex and the third sidewall can be substantially concave. For example, the container body can have a substantially ovular cross-section with major and minor axes. The inlet and the outlet can be aligned along the minor axis. The cap can be configured to couple to the container body via a snap-fit.

In one embodiment, a container for use in a beverage system is provided. The container includes a container body defining an interior hollow chamber and a cap covering the opening in the container body. The container body can have an opening leading to the interior hollow chamber. The cap can have an inlet port, an outlet port, and a collar positioned around the inlet port. The inlet port can have an inlet valve seated therein and can be movable between a closed configuration for preventing passage of fluid there through, and an open configuration for allowing passage of fluid there through. The outlet port can have an outlet valve seated therein and movable between a closed configuration for preventing passage of fluid there through, and an open configuration for allowing passage of fluid there through. The collar can be positioned around the inlet port and can have an inner surface with at least a portion configured to circumferentially sealingly engage a seal having an outer diameter in a range of about 7 mm to 8 mm.

The container can vary in a number of ways and may include any of the following features, alone or in combination. For example, the body can include an end face having the inlet and outlet ports therein, and a skirt extending around the interface portion and defining a sidewall of the body. In some aspects, the skirt can have a substantially triangular shape. In other aspects, the collar can project outward from the end face. For example, the collar can be substantially cylindrical. For example, the inlet valve and the outlet valve each can include a cross-shaped slit configured to enable fluid flow therethrough. For example, the cap can include a closure pivotally coupled thereto and movable between an open position and a closed position. The closure can be configured to close off the inlet valve and the outlet valve in the closed position. In some aspects, the cap can include at least one closure retention feature on an external

surface thereof, and the at least one closure retention feature can be configured to couple to the closure to retain the closure in the open position.

In another embodiment, a container for use in a beverage system is provided. The container can include a container body defining an interior hollow chamber and a cap coupled to the opening of the container body. The cap can have an inlet valve that is sealed to retain the fluid additive within the interior hollow chamber and that is configured to open to allow gas to be injected into the interior hollow chamber, and an outlet valve that is sealed to retain the fluid additive within the interior hollow chamber and that is configured to open when a pressure within the interior hollow chamber exceeds a threshold pressure to allow fluid additive within the container body to flow through the outlet valve. The inlet valve can have a generally cylindrical shape and the outlet valve can have a generally cylindrical shape. A diameter of the outlet valve can be in a range from about 7 mm to 13 mm.

The container can vary in a number of ways and may include any of the following features, alone or in combination. For example, the cap can include a closure pivotally coupled thereto and movable between an open position and a closed position. The closure can be configured to close off the inlet valve and the outlet valve in the closed position. For example, the container body can have a substantially ovalar cross-section including a major axis about a first width and a minor axis about a second width. In some aspects, the inlet port and the outlet port can align with the minor axis of the container body. In other aspects, the cap can include at least one orientation element configured to orient the cap relative to the container body. For example, the inlet valve and the outlet valve each can include a cross-shaped slit configured to enable fluid flow therethrough.

In one embodiment, a flow control assembly is provided. The flow control assembly can include a cap having a flow control system with an inlet port having an inlet valve and an outlet port having an outlet valve. The flow control system can achieve a Dosing Accuracy (DA) of about 100 or less according to the following formula:

$$DA = \left[\frac{(Po - Pc)}{(Vd - Ls)} \right]$$

Po is a pressure to open the outlet valve (mmH₂O), Pc is a pressure to close the outlet valve (mmH₂O), Vd is a diameter of the outlet valve (mm), and Ls is a length of the valve opening (mm).

The flow control assembly can vary in a number of ways and may include any of the following features, alone or in combination. For example, the flow control system can achieve a DA of between about 40 and 70. For example, the flow control system can achieve a DA of about 55. For example, the pressure to open the inlet valve (Po) can be greater than about 100 mmH₂O. For example, the pressure to open the inlet valve (Po) can be greater than about 400 mmH₂O. For example, the diameter of the outlet valve (Vd) can be between about 5 mm and 15 mm, and in certain embodiments can be about 9.5 mm. For example, the length of the valve opening (Ls) can be between about 1 mm and 5 mm, and in certain embodiments can be about 3.7 mm.

In other embodiments, the cap can include a sidewall defining a cavity configured to receive a neck of a container. The cap can include an end wall having the inlet port and an outlet port formed therein. In some aspects, the inlet port and

the outlet port each can include a cylindrical collar having the inlet valve and the outlet valve disposed therein, respectively. For example, the flow control assembly can include a container body defining an interior hollow chamber. The container body can have an opening leading to the interior hollow chamber, and the cap can be configured to couple to the opening of the container body to seal fluid within the interior hollow chamber. In some aspects, the inlet valve can be configured to allow a gas to be injected into the interior hollow chamber, and the outlet valve can be configured to open to allow fluid to flow out of the interior hollow chamber when a pressure within the interior hollow chamber exceeds the pressure to open the outlet valve (Po).

In another embodiment, an ingredient container for use in a beverage carbonation system is provided. The ingredient container can include a container body defining an interior hollow chamber and an opening leading to the interior hollow chamber, and a cap coupled to the opening. The container body can have a cross-section with a major axis defining a width that is greater than a minor axis defining a depth. The cap can have an inlet that can be configured to retain fluid within the container and that can be configured to open to allow gas to be injected into the interior hollow chamber. The cap can have an outlet that can be sealed to retain fluid within the container and that can be configured to open to allow fluid within the container to flow out through the outlet valve. The inlet and the outlet can be aligned along a first axis that extends parallel to the minor axis of the container body.

The container can vary in a number of ways and may include any of the following features, alone or in combination. For example, the first axis can extend substantially perpendicular to the major axis of the container body. For example, the cap can have an irregular shape. For example, the cap can have a substantially triangular outer perimeter. For example, the cap can have a major axis and a minor axis, and the first axis can extend along the minor axis of the cap. For example, the cross-section of the container body can be ovalar.

In another embodiment, an ingredient container is provided. The ingredient container can include a container body defining an interior hollow chamber and having an opening leading to the interior hollow chamber, and a cap positioned over the opening in the container body. The cap can have an irregular shape with a major axis and a minor axis, and the cap can include an inlet port and an outlet port positioned along the minor axis.

The ingredient container can vary in a number of ways and may include any of the following features, alone or in combination. For example, the inlet and outlet port can be positioned along an axis that extends substantially perpendicular to the major axis of the cap. For example, the cap can have a generally triangular cross-sectional shape. For example, the cap can have an outer perimeter with first, second, and third sides, and the first side can be longer than the second and third sides. In some aspects, the inlet and outlet valves can be positioned along an axis extending substantially perpendicular to the first side. For example, the cap can have a base wall having the inlet and outlet ports therein, and a sidewall extending around an outer perimeter of the base wall. The sidewall can have a height that varies around the outer perimeter. For example, the container body can have a cross-section with a major axis defining a width that is greater than a minor axis defining a depth, and the cap major axis can be aligned with the major axis of the container body.

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In another embodiment, an ingredient container is provided. The ingredient container can include a container body having a hollow interior and an opening leading into the hollow interior, and a cap positioned over the opening in the container body and including an inlet port and an outlet port. A cross-section of the cap can extend substantially perpendicular to a central axis of each of the inlet port and the outlet port can have a shape that is a substantially circular triangle.

The ingredient container can vary in a number of ways and may include any of the following features, alone or in combination. For example, the cap can have an outer sidewall defining the shape of the cross-section and can have first, second, and third walls. In some aspects, the first wall can have a length that is greater than a length of each of the second and third walls. In other aspects, the first wall can be substantially planar, and the second and third walls can be convex. For example, the cap can have a base wall with the inlet and outlet ports therein, and an outer sidewall surrounding the base wall. The outer sidewall can have first and second shoulders projecting upward from the base wall. In some aspects, the cap can include a base wall having the inlet and outlet ports formed therein, and the base wall can include a circular cavity formed therein at a mid-portion thereof. In some variations, inlet and outlet ports can be positioned within the circular cavity.

In another embodiment, an ingredient container is provided. The ingredient container can include a container body having an opening leading into a hollow interior, and a cap covering the opening. The cap can include a base having an inlet port and an outlet port formed therein, and a sidewall extending around the base and defining an outer perimeter of the cap body. The sidewall can include first and second shoulders extending upward from the base on opposed sides of the inlet and outlet ports. The first shoulder can have a first inner surface and the second shoulder can have a second inner surface. The first and second inner surfaces each can have a detent therein configured to receive a corresponding protrusion in a carriage assembly of a beverage carbonation system.

The closure can vary in a number of ways. For example, the detent can include an opening formed through the first and second inner surfaces. In some aspects, the opening can be generally rectangular. For example, the closure can include a lid coupled to the cap body. The lid can be movable between an open position spaced a distance from the inlet and outlet, and a closed position in which the lid covers the inlet and outlet. For example, the sidewall can have a generally triangular cross-sectional shape. For example, the first and second inner surfaces can be substantially planar. For example, the first shoulder can have a first outer surface opposite the first inner surface, and the second shoulder can have a second outer surface opposite the second inner surface. The first and second outer surfaces can be convex. For example, the base can include a circular recess formed therein and can have the inlet and outlet port position therein.

In another embodiment, a carbonation system is provided. The carbonation system can include a housing having at least one movable carriage with a cavity therein, and a container having a hollow body and a cap coupled to the hollow body. The cavity can include at least one spring-biased projection. The cap can include a base with inlet and outlet ports, and a sidewall extending around the base and having first and second shoulders, and at least one detent formed on an inner facing surface of at least one of the first and second shoulders. The at least one detent can be

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configured to receive the at least one projection in the carriage when the container is disposed within the cavity in the carriage.

The carbonation system can vary in a number of ways. For example, the at least one projection and the at least one detent can be configured to produce an audible click when the container is inserted into the cavity in the carriage. For example, the at least one projection can include first and second projections positioned within the cavity, and the at least one detent can include first and second detents formed on the inner facing surface of the first and second shoulders, respectively. For example, the inner facing surface of the first and second shoulders can extend substantially perpendicular to the base. For example, the sidewall can have a substantially triangular cross-sectional shape.

In another embodiment, a carbonation system is provided. The carbonation system can include a housing having at least one movable carriage with a cavity therein, and a container having a hollow body and a cap coupled to the hollow body. The cavity can include first and second spring-biased projections. The cap can include inlet and outlet ports, and the cap can have first and second detents formed therein and configured to receive the first and second projections in the carriage when the container is disposed within the cavity in the carriage. The first and second projections and the first and second detents can be configured to produce an audible click when the container is inserted into the cavity in the carriage.

The carbonation system can vary in a number of ways. For example, the cap can include a base having the inlet and outlet ports therein, and a sidewall can extend around the base and can include first and second shoulders. The first and second detents can be formed in the first and second shoulders, respectively. In some aspects, the first and second shoulders can have first and second inner facing surfaces with the first and second detents formed therein, and the first and second inner facing surfaces can extend substantially perpendicular to the base. For example, the cap can have a substantially triangular cross-sectional shape.

In another embodiment, a container is provided. The container can include a container body having an opening extending into a hollow interior, and a cap extending across the opening. The cap can have an inlet port with an inlet valve configured to couple to a fluid source such that fluid can be delivered through the inlet valve to pressurized the hollow interior of the container body, and an outlet port with an outlet valve. The outlet valve can have a cracking pressure at which the outlet valve is configured to move from a closed configuration to an open configuration to dispense fluid from the hollow interior, and a closing pressure at which the outlet valve is configured to move from the open configuration to the closed configuration to prevent fluid from passing therethrough. The cracking pressure can be greater than the closing pressure.

The container can vary in a number of ways. For example, a difference between the cracking pressure and the closing pressure can be in a range of about 300 mmH₂O to 400 mmH₂O. For example, a difference between the cracking pressure and the closing pressure can be about 340 mmH₂O. For example, the cracking pressure can be greater than about 600 mmH₂O or less than about 400 mmH₂O. For example, the inlet valve and the outlet valve each can include a cross-shaped slit configured to enable fluid flow therethrough.

In another embodiment, a container is provided. The container can include a container body having an opening extending into a hollow interior, and a cap extending across

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the opening. The cap can have an inlet port with an inlet valve configured to couple to a fluid source such that fluid can be delivered through the inlet valve to pressurized the hollow interior of the container body, and an outlet port with an outlet valve. The outlet valve can have a closed configuration to prevent fluid flow from the hollow interior, and can be movable to an open configuration to dispense fluid from the hollow interior in response to a pressure increase within the hollow interior increase of between about 300 and 380 mmH₂O.

The container can vary in a number of ways. For example, the pressure increase can be about 340 mmH₂O. For example, the outlet valve can have a cracking pressure greater than about 600 mmH₂O. The outlet valve can have a closing pressure less than about 400 mmH₂O. In some embodiments, the inlet valve and the outlet valve can each have a cross-shaped slit configured to enable fluid flow therethrough.

In another embodiment, a container is provided. The container can include a container body defining a hollow interior, and a cap. The cap can have an inlet port with an inlet valve seated therein and movable between a closed configuration for preventing passage of fluid there through, and an open configuration for allowing passage of fluid there through. The cap can also have an outlet port having an outlet valve seated therein and movable between a closed configuration for preventing passage of fluid there through, and an open configuration for allowing passage of fluid there through. The outlet valve can have a configuration that will dispense a predetermined amount of fluid in a range of 1.6 mL to 2.0 mL in response to a dose of gas being pumped into the container for a period of 140 ms.

The container can vary in a number of ways. For example, the predetermined amount of fluid can be 1.8 mL. For example, the inlet valve and the outlet valve each can include a cross-shaped slit configured to enable fluid flow therethrough. For example, the outlet valve can have a cracking pressure at which the outlet valve is configured to move from a closed configuration to an open configuration to dispense fluid from the hollow interior, and can have a closing pressure at which the outlet valve is configured to move from the open configuration to the closed configuration to prevent fluid from passing therethrough. The cracking pressure can be greater than the closing pressure. For example, the predetermined amount of fluid is proportional to a difference between the cracking pressure and the closing pressure.

The details of one or more variations of the subject matter described herein are set forth in the accompanying drawings and the description below. Other features and advantages of the subject matter described herein will be apparent from the description and drawings, and from the claims.

DESCRIPTION OF DRAWINGS

These and other features will be more readily understood from the following detailed description taken in conjunction with the accompanying drawings, in which:

FIG. 1 is a front view of one embodiment of a beverage dispensing system;

FIG. 2 is a rear perspective view of the beverage dispensing system of FIG. 1 with various housing components removed;

FIG. 3 is a front perspective view of one embodiment of a housing portion and carriage assembly for use with a beverage dispensing system;

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FIG. 4 is a perspective view of a carriage assembly of FIG. 3 having the housing portion removed;

FIG. 5A is a top view of a carriage used with the carriage assembly of FIG. 3;

FIG. 5B is a cross-sectional view of the carriage of FIG. 5A;

FIG. 6 is a bottom view of the carriage of FIG. 5A;

FIG. 7 is a perspective view of an ingredient container according to an embodiment;

FIG. 8A is a cross-sectional view of the ingredient container of FIG. 7;

FIG. 8B is a partial cross-sectional view of the ingredient container of FIG. 7;

FIG. 9 is an exploded view of the ingredient container of FIG. 7;

FIG. 10A is a perspective view off a container body of the ingredient container of FIG. 7;

FIG. 10B is a top view of the container body of FIG. 10A;

FIG. 11A is a perspective view of a lid of the ingredient container of FIG. 7;

FIG. 11B is a top view of the lid of FIG. 11A;

FIG. 12A is a perspective view of an outlet valve of the ingredient container of FIG. 7;

FIG. 12B is a cross-sectional view of the outlet valve of FIG. 12A during a dispensing process;

FIG. 12C is a cross-sectional view of the outlet valve of FIG. 12A during a dispensing process;

FIG. 12D is a cross-sectional view of the outlet valve of FIG. 12A during a dispensing process;

FIG. 12E is a cross-sectional view of the outlet valve of FIG. 12A during a dispensing process;

FIG. 12F is a cross-sectional view of the outlet valve of FIG. 12A during a dispensing process;

FIG. 13 is a rear perspective view of the lid of FIG. 11A;

FIG. 14 is a perspective cross-sectional view of the lid of FIG. 11A;

FIG. 15 is a partial perspective rear view of the lid of FIG. 11A having a lid cover in a closed position;

FIG. 16 is a perspective bottom view of the lid of FIG. 11A;

FIG. 17 is a perspective view of the carriage assembly of FIG. 3 having the ingredient container of FIG. 7 loaded therein;

FIG. 18 is a perspective view of the carriage assembly and ingredient container of FIG. 17 having a housing removed;

FIG. 19 is a perspective view of the carriage of FIG. 5A having the ingredient container of FIG. 7 loaded therein;

FIG. 20 a cross-sectional view of the carriage and ingredient container of FIG. 19;

FIG. 21 is a partial cross-sectional perspective view of the container and ingredient container of FIG. 19; and

FIG. 22 is a bottom view of a carriage assembly according to some embodiments showing a relative position of ingredient container outlets and a fluid outlet in relation to variously-sized drinkware.

It is noted that the drawings are not necessarily to scale. The drawings are intended to depict only typical aspects of the subject matter disclosed herein, and therefore should not be considered as limiting the scope of the disclosure.

DETAILED DESCRIPTION

Certain illustrative embodiments will now be described to provide an overall understanding of the principles of the structure, function, manufacture, and use of the devices and methods disclosed herein. One or more examples of these embodiments are illustrated in the accompanying drawings.

Those skilled in the art will understand that the devices and methods specifically described herein and illustrated in the accompanying drawings are non-limiting illustrative embodiments and that the scope of the present invention is defined solely by the claims. The features illustrated or described in connection with one illustrative embodiment may be combined with the features of other embodiments. Such modifications and variations are intended to be included within the scope of the present invention.

Further, in the present disclosure, like-named components of the embodiments generally have similar features, and thus within a particular embodiment each feature of each like-named component is not necessarily fully elaborated upon. Additionally, to the extent that linear or circular dimensions are used in the description of the disclosed systems, devices, and methods, such dimensions are not intended to limit the types of shapes that can be used in conjunction with such systems, devices, and methods. A person skilled in the art will recognize that an equivalent to such linear and circular dimensions can easily be determined for any geometric shape.

In general, ingredient containers for use with beverage dispensers and carriages for receiving ingredient containers are provided. In one embodiment, an ingredient container is provided that can contain an additive for use in a beverage dispensing process. The ingredient container can have a hollow container body with an opening and a lid coupled to the container body. The lid can include a lid base configured to couple to the container body over the opening, and the lid base can have an inlet and an outlet therein. In certain embodiments, the lid can further include a lid cover configured to selectively close the inlet and the outlet, thereby sealing a hollow interior of the container body. The inlet and the outlet can each have a seal disposed therein that is configured to open in the presence of a pressure differential between an interior and an exterior of the ingredient container in an attempt to eliminate the pressure differential. The ingredient container can be shaped and designed to correspond to a carriage located on a beverage dispensing device. The carriage can have complimentary features to receive and retain the ingredient container, and when retained, the ingredient container can be employed by a beverage dispensing device for use in the creation of customized beverages.

Methods of dispensing the additive stored within the ingredient container can vary. In some embodiments, the ingredient container is pressurized with a gas, such as air, to cause the outlet to open and dispense the stored additive. When the ingredient container is properly seated and retained by a carriage, a gas line fluidly coupled to a pump can receive the inlet of the ingredient container in order to seal around the inlet in preparation for the introduction of gas into the ingredient container during a dispensing procedure. Gas can be pumped by the pump, through the gas line, through the inlet seal, and into the hollow interior of the ingredient container. The resulting increase in internal pressure can cause the outlet seal to open and dispense an amount of the additive proportional to the amount of gas introduced through the inlet.

FIGS. 1-2 illustrate a beverage dispensing system 10 according to one embodiment. The beverage dispensing system 10 can be used to create and dispense customized beverages for a user, based on desired characteristics of the beverage. The illustrated beverage dispensing system 10 generally includes a housing 12 having a fluid reservoir 14 and a carbonation assembly 16. A carriage assembly 18 can be included on and/or coupled to the beverage dispensing

system 10, and it can receive one or more ingredient containers 20 to be used in the creation of beverages. The ingredient containers 20 can include one or more additives (e.g., a flavorant, a vitamin, a food dye, etc.) to be included in a created beverage as desired.

During a beverage dispensing process, a user can actuate inputs located at a user interface 22 in order to select specific characteristics of the desired beverage, such as volume, carbonation level, specific additives, and additive amount. If the user selects inputs to indicate that the beverage is carbonated, water can be fed from the fluid reservoir 14 and into the carbonation assembly 16, and carbon-dioxide can be fed from a canister 24 and into the carbonation assembly 16 to produce carbonated water. If the user selects inputs to indicate that one or more additives should be added to the beverage, the beverage dispensing system 10 can dispense the additive from the one or more ingredient containers 20 coupled to the system. The beverage can be dispensed into a container, such as a drinking glass 26.

FIGS. 3-6 illustrate one embodiment of a carriage assembly 100 which can be coupled to and/or retained with or within a beverage dispensing device, such as beverage dispensing device 10. In the illustrated embodiment, the carriage assembly 100 is contained within a carriage housing 100A. The carriage assembly 100 can include one or more carriages 101, which can each seat and retain one or more ingredient containers (not shown) for use in a beverage dispensing process. Although the carriage assembly 100 is shown having two separately movable carriages 120, a different number of carriages 120 are contemplated herein as well. For example, the carriage assembly can be in the form of a single movable carriage having multiple cavities with each cavity configured to receive an ingredient container. Ingredient containers and their retention within the carriage assembly 100 will be described in greater detail below.

FIG. 4 illustrates the carriage assembly 100 separated from the carriage housing 100A. The illustrated carriage assembly 100 generally includes a left carriage 120L and a right carriage 120R (collectively carriages 120) coupled to a carriage base 110. The carriage base 110 can have a variety of forms, which may depend upon the form of the carriage housing 100A containing the carriage base 110. As illustrated, the carriage housing 100A and the carriage base 110 have a substantially cylindrical form. The carriage base 110 can include cutouts and/or slots for seating and receiving various components including, for example, the carriages 120 and a fluid outlet 114. The carriages 120 can be coupled to be carriage base 110 in a variety of ways, for example, the carriages 120 can be pivotally hinged to the carriage base 110 such that the carriages can pivot downward in order to facilitate loading one or more ingredient containers. The left carriage 120L is illustrated in FIGS. 3 and 4 in an upward position, while the right carriage 120R is pivoted downward to a downward position. The carriages 120 can be coupled to the carriage assembly 100 in other ways, such as via a sliding connection, a stationary connection, etc., or they can be coupled directly to a beverage dispensing device. The illustrated carriage base 110 further includes lift assists 116, which can be coupled to a rear region of the carriages 120. The lift assists 116 can include a biasing feature such as a spring, such that each of the coupled carriages are biased to the upward position. A micro-switch 112 (also referred to as left micro-switch 112L and right micro-switch 112R) can be located above each of the carriages 110, which will be discussed in more detail below.

FIGS. 5 and 6 depict a single carriage 120 in more detail. Features described as applying to one carriage can be

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applied to all carriages. As shown, the carriage **120** has a generally rectangular carriage body **122** with a rounded front face **124** that can be shaped to conform with an overall contour of the carriage housing **110A**. A handle **128** can extend from the front face **124** to provide a grasping surface to enable the carriage **120** to be easily pivoted, such as when an ingredient container is placed into or removed from the carriage assembly **120**. While the handle **128** is shown in the form of a protruding lip or ledge, the handle **128** can take on various forms and can include protrusions of other shapes as well as recesses within the carriage body **122** itself. The carriage **120** can further include a pivot axis **126** located near a rear of the carriage body **122**, as introduced above, for allowing the carriage **120** to pivot relative to the carriage body **122**.

An upper portion of the carriage body **122** can include a carriage face **130**, as best shown in FIGS. **5A** and **5B**. In an exemplary embodiment, the carriage face **130** is shaped to receive and retain a complimentary ingredient container for use during a beverage dispensing process. The carriage face **130** can include a variety of indentations, protrusions, flat areas, and rounded areas to fully receive ingredient containers of any shape or size, as well as to ensure that an ingredient container is properly seated and coupled to the system. In the illustrated embodiment, the carriage face **130** is in the form of a generally triangular recess with rounded corners, e.g., a rounded triangle. One side can include a rectangular cutout **131** extending therefrom. In certain embodiments, the cutout **131** can be formed in the longed side of the triangle, and it can be located closest to a mid-portion of the carriage assembly **100**. The central region **132** of the carriage face **130** can include a raised platform having a variety of features thereon. As shown, the central region **132** is raised such that a peripheral channel **133** is defined within the carriage face **130**. The central region **132** can include a carriage inlet **134** and a carriage outlet **136**, which can be configured to align with and couple to an inlet and an outlet of an ingredient container, respectively. The illustrated carriage inlet **134** and the carriage outlet **136** have a substantially round form defining a central opening **134A**, **136A**. The central openings **134A**, **136A** can pass entirely through the carriage **120**. The inlet and outlet receivers **134**, **136** can be made from a variety of materials. For example, one or both of the inlet and outlet receivers **134**, **136** can be made from a plastic, a resin, a rubber, a metal, or a composite thereof. In certain embodiments, for example, one or more of the inlet and outlet receivers **134**, **136** can be made from a rubber or rubber-like material such that an air-tight seal is created between the carriage face and a seated ingredient container, as discussed further below.

A space around the carriage inlet **134** and the carriage outlet **136** can be recessed into the central region **132**, thereby defining the overall form of both the carriage inlet **134** and the carriage outlet **136**. In the illustrated embodiment, this space, also called the central recess **137**, takes the form of a substantially “figure-8” shape, with the inlet and outlet receivers **134**, **136** being positioned within each opening of the “figure-8.” The central region **132** can also include one or more flanking protrusions **138** disposed proximate to the central recess **137**. The flanking protrusions **138** can be informed by the shape of other features found in the carriage face **130**, or they can have independent designs. In the illustrated embodiment, the flanking protrusions **138** are shaped to extend into complimentary recesses on an ingredient container to assist in the retention thereof. As shown in FIGS. **5A** and **5B**, the carriage face **130** includes a pair of similar flanking protrusions **138**, which take the

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form of “bat wings” that follow the “figure-8” contour of the central recess **137**. In particular, the illustrated flanking protrusions **138** have an outer sidewall that is convexly curved along its length and two inner sidewalls that are each convexly curved to follow the contours of the inlet and outlet receivers **134**, **136**. The sidewalls of the flanking protrusions **138** can taper in a direction leading away from the carriage face **130**, as shown, such that the tip portion is generally smaller in size than the base portion of each protrusion **138**. In other embodiments, the protrusions **138** may not flare at all.

The peripheral channel **133** can further include one or more features to assist in the retention of an ingredient container. As will be discussed in more detail below, each peripheral channel **133** can have a shape configured to complement a shape of the container such that two shoulders on the container, as well as other portions of the container, can be received therein. In the illustrated embodiment, the peripheral channel **133** includes two generally rounded triangular areas and an elongated slot extending therebetween. The channel **133** is defined by the shape of the center region **132**, which is generally square with rounded corners, in combination with the shape of the generally triangular recess in the carriage face **130**. The peripheral channel **133** can also include one or more retainers **139** protruding from a sidewall of the center region **132** outward into the peripheral channel **133**. The retainers **139** can be spring-biased outward, such that during a retention process the retainers **139** can be forced inward by the container before springing back outward to engage a corresponding recess in the ingredient container. The retention process will be described in greater detail below.

FIG. **5B** illustrates the relative heights of the carriage **120**, including the carriage inlet **134**, the carriage outlet **136**, and the flanking protrusions **138**. As shown, the carriage outlet **136** has a height that is greater than a height of the carriage inlet **134**. The retainers **139** can be seen located within the peripheral channel **133**, which is set below the elevated center region **132**. While the carriage face **130** is described and shown as having certain areas recessed and other areas protruding, carriage faces with the opposite features are contemplated herein as well, i.e., all protrusions are recesses and all recesses are protrusions. Further, carriage faces are also contemplated that may have only a portion of the features interchanged, such that only one or a few protrusions are recesses and/or only one or a few recesses are protrusions. Other shapes and configurations are also contemplated.

FIG. **6** depicts an underside of the carriage **120**, according to some embodiments. The underside of the carriage **120** is positioned on the opposite side of the carriage inlet **134** and the carriage outlet **136**, and it includes central holes **134A**, **136A**, which, as introduced above, can pass through the carriage **120**. In operation, the central hole **134A** of the carriage outlet **136** can be coupled to a gas line **140**. The gas line **140** can be coupled at an opposite end to an air pump (not shown), which can be used to introduce air or another gas into a seated ingredient container. The resulting increase in pressure can cause the seated ingredient container to dispense a stored additive through the central hole **136A** of the carriage outlet **136**. In systems with more than one carriage, one or more pumps can be used to introduce gas to a seated ingredient container. In some variations, each carriage can have its own pump fluidly coupled thereto via a gas line or similar setup. In other embodiments, the gas line **140** can be coupled to the carbonation source, which can be used to supply gas to the container for ejecting additive.

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FIGS. 7-16 illustrate an exemplary embodiment of an ingredient container 200. The ingredient container 200 can generally include a lid 210 coupled to a container body 250 which can be configured to contain an additive (e.g., a flavorant, a supplement, a vitamin, a coloring agent, etc.) to be used in the creation of beverages. The additive can be in the form of a fluid, a solid, a powder, a gel, a syrup, or any other form. The ingredient container 200 can come in a variety of sizes. For example, the ingredient container 200 can have an overall height between about 55 mm and 60 mm, and in some embodiments can be about 56.9 mm. The ingredient container 200 can have a maximum width between about 55 mm and 65 mm, and in some embodiments, the maximum width can be about 59.5 mm. The lid 210 can have a depth between about 38 mm and 42 mm, and in some embodiments can be about 39.6 mm. The container body 250 can have a depth between about 38 mm and 42 mm, and in some embodiments can be about 39.5 mm. For example, the ingredient container 200 can have a volume between about 50-90 mL, and in some variations can have a volume of about 70 mL.

The ingredient container 200 can store the additive inside, and, as part of a beverage creation process, receive a measured volume of gas (e.g., air, carbon-dioxide, etc.) through an inlet 224 resulting in an increased internal pressure. The increase in internal pressure within the container 200 can result in an outlet 226 emitting a tailored amount of the additive as a consequence of eliminating or reducing the newly-created pressure differential across the outlet.

The illustrated container body 250 has a generally oblong, ovular form similar to a race-track configuration, as seen in FIGS. 10A and 10B. While the container body 250 is shown as having a specific form, the container body 250 can take on a variety of forms. This oblong ovular form can include a minor width W1 about a shorter dimension of the container body 250 and a major width W2 about a longer dimension of the container body 250. Similarly, the oblong, ovular form can have a minor axis A1 extending centrally along the minor width W1, and the oblong, ovular form can have a major axis A2 extending centrally along the major width W2. As will be discussed in more detail below, the shape of the container body can aid in allowing multiple containers to be positioned closer to one another within the beverage system, thus allowing the outlets 226 to be positioned closer for dispensing an additive.

The container body 250 can include a base 252, a sidewall 254 extending upwardly from the base 252, and a top 256, which together can define an interior space to store the additive. In some embodiments, the base 252 can include an ovular recess 253 as shown in FIG. 8B. The ovular recess 253 can provide increased structural integrity to the container body 250 during storage, transit, operations, etc., and it can also provide an area for increased engagement, such as by a user and/or by a beverage dispensing device (e.g., beverage dispensing system 10).

The sidewall 254 can extend upward from the base 252 to maintain a substantially constant cross-section. The sidewall 254 can include first and second side faces 254A, 254B, which can be substantially planar, and first and second convexly curved faces 254C, 254D extending between the first and second side faces 254A, 254B. A series of channels 255 can run vertically on the first and second side faces 254A, 254B, substantially parallel to each other. The channels 255 can operate similarly to the ovular recess 253, in that they may provide for increased structural integrity, and/or they may provide an area of increased engagement

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between the container body 250 and a beverage dispensing device (e.g., beverage dispensing system 10). They can also aid in gripping the container. In certain embodiments, a carriage assembly (e.g., carriage assembly 100) can have complimentary components to be received by the channels 255 in order to aid in retention of the ingredient container 200.

The top 256 sits upon the sidewall 254, and it can include a shoulder 258 and a neck 260. The shoulder 258 can have a gradual slope upward toward the neck 260, which can be centrally disposed on the top 256 and can be a round, substantially vertical portion of the container body 250. The neck 260 can define the opening 262 leading to the interior of the container body 250. A circumferential flange 264 can extend around the neck 260 and can provide a coupling point for the lid 210, such as with a snap-fit. In some embodiments, the circumferential flange 264 can be replaced by threads to provide threaded connection with the lid 210. A pair of orientation protrusions 266 can be disposed on opposite sides of the neck 260. These protrusions 266 can vary in shape or number, and they can function to align with complimentary features on the lid 210 to ensure that the lid 210 is properly oriented on the container body 250.

FIGS. 11A-16 depict the lid 210 and elements thereof, separated from the container body 250. The illustrated lid 210 has a substantially rounded triangular shape and includes a lid base 220 and a lid cover 240 coupled to the lid base 220. The lid cover 240 can be used to close the inlet and outlet, and in turn the container body 250. The triangular shape can be defined by a perimeter having first, second, and third sides, with the first side being longer than each of the second and third sides.

The lid base 220 can include a skirt 222 located at a lower perimeter thereof and having a curved shaped to conform with the shoulder 258 of the container body 250. The skirt 222 can include a front recess 223, which can be shaped to allow a portion of the lid cover 240 to extend outward beyond the skirt 222 when the lid 210 is in the closed position to enable grasping of the lid cover 240 to case opening and closing of the lid cover 240 relative to the lid base 220. The lid base 220 can include an inlet 224 and an outlet 226, which lead respectively to and from the interior of the container body 250. The inlet 224 can include an inlet collar 224A flanking an inlet orifice 224B, while the outlet 226 can include an outlet collar 224A flanking an outlet orifice 224B. In the illustrated embodiment, the inlet collar 224 has a height that is greater than a height of the outlet collar 226. The greater height of the inlet collar 224 can aid in allowing a seal to be formed between the container inlet 224 and the outlet 136 on the carriage 120.

The container inlet 224 and the outlet 226 can be positioned on the lid base 220 in line with a minor axis B-B of the lid 210 extending along a plane defined by an upper face of the lid base 220, as shown in FIG. 11A. When the lid 210 is coupled to the container body 250, the minor axis B-B can extend parallel to the minor axis A1 of the container body 250, and therefore can extend perpendicular to the major axis A2. In some variations, the entire lid 210 can be substantially symmetrically mirrored about the minor axis B-B. The lid 210 can also have a major axis A-A, as seen at least in FIGS. 11A-11B, which can extend perpendicular to the minor axis B-B.

As shown in FIG. 11B, the inlet 224 and outlet 226 can each have a central longitudinal axis (also called a central axis) with a distance D there between. The central longitudinal axis of each of the inlet 224 and the outlet 226 is coming out of the page in FIG. 11B, but it is shown from a

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side view in FIG. 21. The distance D between each central longitudinal axis can vary. In certain embodiments, the distance D can depend at least partially on the overall dimensions of the lid 210 and/or the sizes of the valves, as discussed further below. For example, in some embodiments, the distance D between the central axes can be between about 9 mm and 15 mm, and more preferably between about 11 mm and 13 mm, and in certain exemplary embodiments the distance D can be about 13 mm.

As further shown in FIG. 11B, the inlet 224 can have a diameter X1, and the outlet 226 can have a diameter X2. The inlet diameter X1 can be between about 6.6 mm and 7.2 mm, and in some embodiments can be about 6.90 mm. The outlet diameter X2 can be between about 6.5 mm and 7.1 mm, and in some embodiments can be about 6.84 mm.

Recesses 228 can flank each side of the inlet 224 and the outlet 226, and the recesses 228 can each be shaped to correspond to protrusions in a carriage (e.g., flanking protrusions 138 on carriage 120). For example, the recesses 228 can be shaped to follow an outer contour of the collars 224A, 226A and can take a “bat wing” form. In particular, similar to the flanking protrusions 138, the recesses 228 can have a radially outward sidewall that is concavely curved along its length and two inner sidewalls that are concavely curved to follow the contours of the inlet and outlet 224, 226. The recesses 228 can take on various other forms as well, and their form may be at least partially dependent upon the placement and form of other components on the lid 210. The recesses 228 can be placed a slight distance apart from the inlet 224 and the outlet 226, thus defining a central pattern 230 located in the space between the collars 224A, 226A and the recesses 228. As best seen in FIG. 11A, the central pattern 230 can take the form of a “figure-8,” however other forms may be present. The illustrated central pattern 230 is shown being flush with the upper surface of the base 220, however the central pattern 230 can protrude above the upper surface or can be recessed below the upper surface. The central pattern 230 can be a protrusion, a recession, or a combination thereof with a portion of the central pattern 230 protruding from the lid 210 and a portion of the central pattern 230 receding into the lid 210. In some variations, the inlet and outlet collars 224A, 226A can contribute to the central pattern 230.

As explained previously with respect to the carriage face 130, although the lid base 220 is described and shown as having certain areas recessed and other areas protruding, lid bases with the opposite features are contemplated herein as well, i.e., all protrusions are recesses and all recesses are protrusions. Further, lid bases are also contemplated that may have only a portion of the features interchanged, such that only one or a few protrusions are recesses and/or only one or a few recesses are protrusions.

The lid base 220 can further include a pair of shoulders 231 formed on opposed sides of the skirt 222 and that extend upward from the lid base 220. Each shoulder 231 can have a shape, such as a rounded triangular shape, that complements a shape of the peripheral channel 133. Each shoulder 231 can also include one or more retention features, which can further assist in retention of the ingredient container 200 within the carriage 120. These features can be in the form of receivers 232 which can receive a complimentary element of the carriage 120, as will be described in more detail below. In the illustrated embodiment, the receivers 232 are each in the form of a substantially square or rectangular recess or cut-out formed in an inward facing sidewall of each shoulder 231.

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As further shown, a rear portion of the lid base 220 can include a rear wall 233 which can extend between the shoulders 231. The lid cover 240 can be coupled to the rear wall 233, as will be discussed in more detail below.

Referring again to the inlet 224 and the outlet 226, as previously explained the inlet 224 can include an upwardly extending inlet collar 224A flanking an inlet orifice 224B, and the outlet 226 can include an upwardly extending outlet collar 226A flanking an outlet orifice 226B. Although the inlet collar 224A and the outlet collar 226A are shown in a circular form, the inlet and outlet collars 224A, 226A can take on a number of shapes, including various geometric shapes, e.g., a triangle, a star, etc., as well as fanciful and/or irregular shapes, e.g., a letter, a logo, etc. The form of the inlet and outlet collars 224A, 226A can be the same or different. As shown in FIG. 9, the inlet 224 can include an inlet valve frame 224C and an inlet valve 224D, and the outlet 226 can include an outlet valve frame 226C and an outlet valve 226D. Generally, each of the inlet valve 224D and the outlet valve 224D can be respectively seated within the inlet valve frame 226C and the outlet valve frame 226C. The inlet valve frame 224C and the outlet valve frame 226C can be affixed to the underside of the lid 210 beneath the inlet 224 and the outlet 226 respectively. In other embodiments, the inlet valve frame 224C and the outlet valve frame 226C can be formed from a single frame component.

FIG. 12A depicts one embodiment of an outlet valve 226D in more detail. While description is made with respect to the outlet valve 226D, similar features are applicable to the inlet valve 224D. Additionally, where options are provided for aspects of the outlet valve 226D, actual aspects may not always be the same between the inlet valve 224D and the outlet valve 226D. The illustrated outlet valve 226D is configured to open to dispense an additive therefrom during a beverage dispensing process. While the outlet valve 226D is depicted as being round or substantially circular, the outlet valve 226D can vary in form to have any number of regular or irregular shapes. In general, the outlet valve 226D can include a flange 226E configured to hold an outlet valve head 226F within the outlet valve frame 226C. The flange 226E can be connected to the outlet valve head 226F via a roll sleeve 226G. The outlet valve 226D can also vary in size, and the size can depend at least in part on the diameter of the outlet 226 itself. For example, the outlet valve diameter Vd of the outlet valve 226D on the container body 250, i.e., not including the flange 226E, can be between about 8 mm to 12 mm. In some embodiments, the outlet valve diameter Vd can be between about 9 mm and 10 mm. The outlet valve 226D can be in the form of a slit valve having a slit 226H configured to open and allow for the transfer of a material, such as a fluid, therethrough. The slit 226H can have a variety of forms and sizes. For example, as shown in FIG. 12A, the slit 226H has a cross or X shape. The slit 226H can vary in size, but in an exemplary embodiment it can have a slit length Ls between about 1.5 mm and 5.5 mm. Note the slit length as used herein refers to the length of the longest slit where two or more slits are provided. In some embodiments, the slit length Ls can be between about 1.5 mm and 2 mm, and the outlet valve 226D can open at the cross-shaped slit 226H when subjected to enough pressure, either internally or externally. An opening pressure Po (also called a cracking pressure) of the outlet valve 226D can vary, and can be dependent upon the material, size, or other details of the outlet valve 226D. For example, in some embodiments, the opening pressure Po can be about 300 mmH₂O or greater, and more preferably about 600 mmH₂O or greater. A closing pressure Pc of the outlet valve 226D can vary as

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well, and can be dependent upon various details of the outlet valve 226D. In some embodiments, the closing pressure P_c can be about 400 mmH₂O or less. In other embodiments, the closing pressure P_c can be about 300 to 400 mmH₂O less than the cracking pressure.

When the outlet valve 226D is subjected to a high enough pressure differential, such as in the build-up to attaining the opening pressure P_o and then subsequent achievement of the opening pressure P_o , the valve 226D can undergo a several-step transformation process before opening at the slit 226H. This transformation process is illustrated in FIGS. 12B-12F. In FIG. 12B, the outlet valve head 226F begins to move downward, subject to some pressure, rolling about the outlet valve sleeve 226G. In FIG. 12C, the outlet valve sleeve 226G is fully unrolled. In FIG. 12D, the outlet valve head 226F begins to flatten, and then at FIG. 12E, the opening pressure P_o is achieved, forcing the slit 226H open and dispensing an additive. When the slit opens, the pressure differential across the valve 226D quickly dissipates, and the valve head 226F can return to its typical position. As a result of this return, in some configurations, the slit 226H can open inwardly, as shown in FIG. 12F, before finally reaching a rest state and returning to the position depicted in FIG. 12A. In other configurations, an internal pressure on the outlet valve 226D can cause the outlet valve 226D, after opening, to return to the state depicted in FIG. 12B, and the outlet valve 226D may never fully return to the state shown in FIG. 12A.

In some embodiments, the inlet valve 224D can be positioned in the same orientation as the outlet valve 226D. In these embodiments, fluid flows through the inlet valve 224D in the opposite direction as the fluid flowing through the outlet valve 226D, i.e., fluid flows into the ingredient container 200 through the inlet valve 224D but flows out of the ingredient container 200 through the outlet valve 226D, all while the inlet and the outlet valves 224D, 226D are positioned in the exact same orientation. As a result, in these embodiments, the inlet valve 224D does not undergo the same series of steps shown in FIGS. 12A-12F when fluid flows therethrough. Instead, the inlet valve begins in the state shown in FIG. 12A and when subjected to pressure great enough to open the inlet valve 224D, the inlet valve 224D merely opens in a manner similar to the state shown in FIG. 12F, but facing the direction shown in FIG. 12A, thus allowing fluid to flow through the opening. Because fluid is flowing through the inlet valve 224D in a direction that is opposite a direction of fluid flowing through the outlet valve 226D, the inlet valve 224D does not undergo the series of steps involving rolling to an expanded state and then opening, as depicted in FIGS. 12B-12E.

As previously indicated, the lid 210 can also include a lid cover 240, shown in FIGS. 11 and 13, which can be connected to the rear wall 233 by various means, including by a hinge 234 (e.g., a living hinge). The lid cover 240 can include an inlet cover 242 and an outlet cover 244, which are sized to respectively close the inlet 224 and the outlet 226 on the lid base 220. Each of the inlet cover 242 and the outlet cover 244 can include respective inlet and outlet cover collars 242A, 244A that are sized to be internally received by the inlet collar 224B and the outlet collar 226B, as seen in the cross-section of FIG. 14. The outlet cover 244 can also include a central plug 244B that is sized to be internally received by the outlet 226 itself. The central plug 244B can operate to prevent premature opening of the outlet valve 226A. The central plug 244B can protrude out from the lid cover 240 beyond the protrusion distance of the outlet cover collar 244A in order to facilitate closure of the outlet 242 when the lid cover 240 is in the closed position.

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In some embodiments, the lid can include features to hold the cover 240 in an open position. For example, as shown in FIG. 15, the lid base 220 can include a back side 235 having a substantially flat central face 236 with a width that is substantially equal to a width of the lid cover 240. One or more lid cover retention features 236 can be located at an upper end of the back side 235 near the hinge 234. These features 236, which can be in the form of cut-outs or recesses, can secure the lid cover 240 when the lid cover 240 is in an open position. As shown, the lid cover 240 can include cover tabs 246 extending from at least one side of the lid cover 240. The cover tabs 246 can extend into the cover retention features 236 to assist in retention of the lid cover 240 in the open position. In the closed position, the inlet and outlet cover collars 242A, 244A, as explained previously, can extend into and frictionally engage the inlet 224 and the outlet 226. This frictional engagement can assist in retention of the lid cover 240 in the closed position. Additionally, the inlet and outlet cover collars 242A can prevent the inlet and outlet valves 224D, 226D from opening prematurely, such as during transportation. For example, the outlet valve 226D can be prevented from rolling about the roll sleeve 226G as illustrated in FIGS. 12B-12D.

FIG. 16 depicts an underside of the lid 210. The lid base 220 can have a divided, arcuate rim 229 that is sized to couple with the neck 260 (not shown) of the container body 250. The arcuate rim 229, can couple to the neck 260 (not shown) of the container body 250 (not shown) via a snap-fit, threads, and the like. For example, the arcuate rim 229 can include an internal lip 229A that is configured to interface with the flange 264 located on the neck 260. This engagement can be seen especially in FIG. 8C. In an exemplary embodiment, the arcuate rim 229 includes a ridge that engages a corresponding feature on the neck 260 to form a snap-fit connection. Depending upon the means by which the arcuate rim 229 affixes to the container body 250, the physical structure of the arcuate rim 229 may change accordingly. While not shown, a seal such as an O-ring can be disposed within the rim 229 to aid in coupling the lid 210 to the container body 250.

As further shown in FIG. 16, an inner surface of the skirt 221 can include one or more orientation channels 227 that can receive the orientation protrusions 266 (not shown) found on the top 256 of the container body 250 to aid in orientation of the lid 210 on the container body 250. As a result, the lid 210 can be limited to mating to the container body 250 in only two orientations. In embodiments where the lid 210 includes one or more recesses 228 to facilitate coupling with a carriage (e.g., carriage 120), the recesses 228 can extend downward from the underside of the lid 210 in between the inlet and outlet valve frames 224C, 226C, and the arcuate rim 229. In some variations where the recesses 228 are at least partially defined by the shape of the inlet 224, the outlet 226, and the overall shape of the lid 210, the recesses 228 can occupy the entirety of the space found between the inlet and outlet valve frames 224C, 226C, and the arcuate rim 229. Essentially, the recesses 228 can change in form depending upon other features located on the lid 210, such as the inlet collar 224A, the outlet collar 226A, the inlet valve frame 224C, the outlet valve frame 226C, the arcuate rim 229, the retention pattern 230, and more. In other variations, the recesses 228 can occupy only a portion of this space.

FIGS. 17-22 depict the ingredient container 200 retained within the carriage assembly 100. FIG. 18 depicts the carriage assembly 100 with the carriage housing 110A (not

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shown) and the right carriage 120R (not shown) removed. With the right carriage 120R removed, the fluid outlet 114 is more visible.

FIGS. 19-22 depict the ingredient container 200 seated within a carriage 120 in greater detail and from various angles to illustrate components on the ingredient container 200 and the carriage 120 coupling together. In order to seat the ingredient container 200, the lid cover 240 can be retained in the open position, as explained above. The carriage 120 can be lowered to expose the carriage face 130, and the ingredient container 200 can be aligned with the carriage face 130 and pressed down so that the carriage inlet 134 engages the inlet 224 and the carriage outlet 136 engages the outlet 226. When seated, the flanking protrusions 138 can extend into the recesses 228, as best shown in FIG. 20. Both the carriage inlet 134 and the carriage outlet 136 can extend respectively into the inlet 224 and the outlet 226, and the inlet collar 224A and the outlet collar 226A can extend circumferentially around the carriage inlet 134 and the carriage outlet 136, as best shown in FIGS. 21 and 22.

While not shown in FIGS. 19-22, in embodiments with a retainer 139, the retainer 139 can also snap into the receivers 232, which may provide an audible signal for a user to know that engagement is successful, such as an audible click when the retainer 139 is engaged and no longer under tension. The rectangular cutout 131 in the carriage face 130 can receive the lid cover 240 when the lid cover 240 is in the fully-opened position, as seen in FIG. 19. Once the container 200 is fully seated, the carriage 120 can be returned to an elevated position. In some embodiments, such movement of the carriage 120 can actuate the corresponding micro-switch 112 and signal to the dispensing system 10 that the container 200 is seated within the carriage assembly 100.

FIG. 21 depicts cross-sectional views of the container 200 seated within the carriage 120. Several distances and dimensions are highlighted relating to the carriage inlet 134, the carriage outlet 136, the inlet 224, and the outlet 226. These distances and dimensions include a distance D between the central longitudinal axis of the inlet 224 and the central longitudinal axis of the outlet 226 and an outlet valve diameter Vd. Also illustrated are a diameter Y1 of the carriage inlet 134, a diameter Y2 of the carriage outlet 136, and an allowable misalignment 291 between the carriage outlet 136 and the outlet 226. The allowable misalignment 291 can define the effective difference in distances between the respective components of the carriage 120 and the container 200, while still enabling a beverage dispensing process to take place. As seen in more detail in FIG. 22, the carriage outlet 136 can include an outlet receiver rim 136A (also known as a seal) that is sized to fit within the outlet 226. The outlet receiver rim 136A and the outlet 226 can together form a sealing surface such that an additive can be dispensed from the outlet 226 during a beverage dispensing process without concern for leaks or inaccurate dosages. If an ingredient container were seated on the carriage 120 and the dimensions of that ingredient container were such that the outlet receiver rim 136A was not properly received in the container outlet 226, then a beverage dispensing process could potentially be compromised. In some aspects, the rim 136A (or seal) can be between about 7 mm to 8 mm in diameter.

In certain embodiments, the distance D between the inlet and outlet can be between about 11 mm and 15 mm, and in some embodiments it can be about 13 mm. Vd can be between about 8 and 11 mm, and in some it embodiments can be about 9.5 mm. Y1 can be between about 7.7 mm and 8.1 mm, and in some embodiments it can be about 7.91 mm.

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Y2 can be between about 7.5 mm and 7.9 mm, and in some embodiments it can be about 7.70 mm. The allowable misalignment 291 can be between about 0.3 mm and 0.6 mm, and in some embodiments it can be about 0.5 mm.

When the ingredient container 200 is properly seated in the carriage assembly 100, a beverage dispensing process can occur using the stored additive. A user can select their beverage preferences, specifying details including volume, carbonation level, additive type, additive amount, and more. When the selections are received by the dispensing system 10, a beverage can be dispensed with the selected characteristics.

If an additive is desired, air or another gas, including carbon dioxide, nitrogen, oxygen, and the like, can be pumped through the gas line 116 and into the interior of the container body 250 through the inlet port 142 in the carriage 120 and through the inlet valve 244D in the container 200. The resulting increase in pressure within the ingredient container 200 can cause the outlet valve 226D to open and additive to dispense through the outlet 226 and the outlet port 244, into a drink container, such as the drinking glass 26 depicted in FIG. 1. In embodiments where the additive is a fluid, the additive can be dispensed at a certain dispensing flowrate F under a certain pressure. For example, in some embodiments, the dispensing flow rate F can be between about 1 mL/sec and 4 mL/sec. In other embodiments, the dispensing flow rate F can be about 2 mL/sec. A base liquid, such as carbonated water, can also be dispensed from the fluid outlet 114 such that the base liquid and the additive combine in the drinking glass 26.

In an exemplary embodiment, the carriage assembly 100 and two ingredient containers 200 can be arranged to minimize a distance between the fluid outlet 114 of the carriage assembly 100 and the outlets 226 of the ingredient containers 200. A bottom perspective of this arrangement is illustrated in FIG. 22. Although the outlets 226 and the fluid outlet 114 can be distinct, the distance between each outlet 226 (and the outlet port 244, in turn) and the fluid outlet 114 can be minimized as a result of the overall carriage assembly 100 configuration. The minimization of distance can arise as a result of the position of each outlet 226 on the respective ingredient containers 200, located on a minor axis B-B (not shown). When the ingredient containers 200 are received in the carriage assembly, the containers 200 can be positioned such that each outlet 226 is centrally located and close to the fluid outlet 114, which can extend between the two carriages 120, as shown above, for example, in FIG. 18.

This minimized distance can allow for a variety of drink containers to be placed beneath the carriage assembly 100 and to receive a beverage while also minimizing splashing and overall mess. For example, several circles indicative of a scale are shown in FIG. 19, and they can represent, in order of smallest to largest, a narrow water bottle circumference α , a highball glass circumference β , a Collins glass circumference γ , a mason jar circumference δ , and a pint glass circumference ϵ . These circumferences are meant to illustrate the variety of drinkware usable with the beverage dispensing system 10 as a result of the arrangement of the containers 200 within the carriage assembly.

During a dispensing procedure, accurate dosages can be important to the creation of a beverage and can affect the quality of the resulting product. This accuracy can be affected by a number of parameters, each introduced and described above, including opening pressure Po, closing pressure Pc, the outlet valve diameter Vd of an outlet valve, and the slit length Ls on the outlet valve.

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Each of these parameters can affect an overall accuracy of the ingredient container **200** during a beverage dispensing process. For example, if the opening pressure P_o and/or the closing pressure P_c are too low, minor fluctuations in the internal pressure of the ingredient container **200** during a dispensing process, such as those associated with normal tolerance levels of the beverage dispensing device **10**, could contribute to inaccurate dispensing of an additive. Conversely, if the opening pressure P_o and/or the closing pressure P_c are too high, the additive could be dispensed in an extreme manner, resulting in excess spray of the additive and also resulting in inaccurate dispensing.

As explained above, the opening pressure P_o is the pressure required to open the outlet valve **226D** and permit fluid to flow therethrough. Once the outlet valve **226D** is open and fluid is dispensed, the built-up pressure will taper off and decrease over time. Eventually, the pressure will reach a value that is too low to keep the outlet valve **226D** open. This lower limit is the closing pressure P_c . The difference ΔP between the opening pressure P_o and the closing pressure P_c can be optimized so as to not be either too great or too small, as this can affect the overall dosing accuracy during flavoring. The overall structure of the outlet valve, including its size, shape, and material, can alter the value of the opening pressure P_o and closing pressure P_c , which can affect performance of the ingredient container **200**. For example, if the difference ΔP is too small, minor fluctuations during a dispensing procedure could cause the outlet valve **226D** to prematurely open or close. If the difference ΔP is too large then the outlet valve **226D** could have trouble closing once opened, which could result in an inability to add small doses of an additive.

If the opening pressure P_o is too high, the dispensing of fluid can become explosive, unmeasured, and/or unpredictable during dispensing, which can result in an overall loss of dosing accuracy. If opening pressure P_o is too low, minor fluctuations or disturbances could lead to leaking and accidental discharge of an additive, which could also result in an overall loss of dosing accuracy. If the closing pressure P_c is too high, especially relative to the opening pressure P_o (which would result in a small difference ΔP), then the window at which the outlet valve **226D** is open would shrink drastically, which can result in a temperamental valve that is only able to open at a small pressure window. Conversely, if the closing pressure P_c is too low, then the outlet valve **226D** will be open for too long as an additive is dispensed, which can also lead to an overall loss of dosing accuracy as the outlet valve **226D** could fail to close in a precise manner, leading to over-dosing of the additive. Accordingly, the opening pressure P_o and closing pressure P_c can be optimized to result in accurate dosing.

Additionally, the outlet valve diameter V_d and the slit length L_s —values affecting the dimensions of the outlet **226** and the outlet valve **226D**—can effect dosing accuracy if they are too large or small. Forcing an additive out of a too-small or too-large slit **226H** or outlet **226** can affect process timing and overall dosing, thereby affecting the accuracy of the dispensing process.

These values can vary depending upon the manufacturing process, materials, quality, etc. of the ingredient container **200**. Together, these values can contribute to a so-called Dosing Accuracy (DA) value, which can be used to rate the quality of an ingredient container **200**. In general, a given outlet valve on the ingredient container **200** can have a maximum potential in terms of a DA value, such as being able to accurately doze an additive, having a low minimum dose threshold for precision dosing, etc. This maximum

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potential can be limited, in some embodiments, by a quality of an inlet valve on the ingredient container **200**. For example, for a given outlet valve, a quality inlet valve will mean that the maximum potential of the ingredient container **200** can be achieved or at least nearly achieved. However, for the same outlet valve, a poor inlet valve can result in a large drop-off in performance from the outlet valve's potential.

The DA value can be expressed by the following formula:

$$DA = \left[\frac{(P_o - P_c)}{(V_d - L_s)} \right]$$

The individual values of these variables can vary, however their relationship according to the DA formula can provide a simple way to compare the quality of valves. Table 1 lays out several example values according to various designs, with each having the same outlet valve diameter D . It has been discovered that, according to the above formula, a valve having a DA value of 100 or less can accurately dose an additive in order to create a consistent beverage product. Each of the provided examples yields a DA value of 100 or less, with the exception of Examples 16 through 21. These examples pertain to valves which do not provide the ability to accurately dose an additive, likely due to the combination of properties of the valve, including the higher slit length L_s and higher difference ΔP between the opening and closing pressures P_o , P_c .

TABLE 1

Examples						
Example	P_o	P_c	V_d	L_s	ΔP	DA
1	483	343	9.5	2.5	140	20
2	483	323	9.5	1.8	160	21
3	1819	1628	9.5	3.7	191	33
4	660	455	9.5	1.8	205	27
5	665	450	9.5	3.7	215	37
6	483	267	9.5	3.7	216	37
7	660	409	9.5	2.5	251	36
8	483	224	9.5	4.7	259	55
9	483	218	9.5	5.1	265	60
10	1448	1163	9.5	1.8	284	37
11	960	640	9.5	2.5	320	46
12	660	318	9.5	3.7	342	59
13	1427	1019	9.5	2.5	409	59
14	848	419	9.5	3.7	429	74
15	660	208	9.5	4.7	452	95
16	660	191	9.5	5.1	469	107
17	1405	724	9.5	3.7	681	117
18	2212	1483	9.5	3.7	729	125
19	4575	3759	9.5	3.7	816	140
20	1379	305	9.5	4.7	1074	226
21	4234	2593	9.5	3.7	1641	282

The DA factor, in some embodiments can be less than 100 according to the above formula, and it could fall more specifically between about 40 and 70. In further embodiments, the DA factor can be about 55. In systems with a DA factor that is less than 100, beverage making processes can accurately dose an additive to within fractions of a mL. For example, an amount of additive, such as a fluid, dispensed during a process can be between about 1.6 mL and 2.0 mL, and in some embodiments can be about 1.8 mL. This volume of fluid can be dispensed after gas is pumped into the container for a predetermined time period, such as about 140 ms. Importantly, the amount of fluid dispensed by a con-

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tainer can be proportional to a difference between the opening and closing pressures of a given valve.

As indicated above, the various properties of the valve can vary. In certain exemplary embodiments, the valve has an opening pressure P_o that is about 300 mmH₂O or greater, and more preferably is about 400 mmH₂O or greater, or even 600 mmH₂O or greater; a closing pressure P_c that is less than the opening pressure P_o but that is about 100 mmH₂O or greater, and more preferably is about 300 mmH₂O or greater, or even, in some embodiments, 400 mmH₂O or greater; a pressure differential (ΔP) that is in range of about 200 mmH₂O to 500 mmH₂O, and more preferably is about 300 mmH₂O to 400 mmH₂O, and even more preferably is about 340 mmH₂O; and an outlet valve diameter V_d in a range of about 5 mm to 15 mm. In certain embodiments, V_d can be about 7 mm to 13 mm, and more preferably about 9.5 mm; a slit length L_s in a range of about 1 mm to 5 mm, and more preferably is about 3.7 mm.

Certain illustrative implementations have been described to provide an overall understanding of the principles of the structure, function, manufacture, and use of the systems, devices, and methods disclosed herein. One or more examples of these implementations have been illustrated in the accompanying drawings. Those skilled in the art will understand that the systems, devices, and methods specifically described herein and illustrated in the accompanying drawings are non-limiting illustrative implementations and that the scope of the present invention is defined solely by the claims. The features illustrated or described in connection with one illustrative implementation may be combined with the features of other implementations. Such modifications and variations are intended to be included within the scope of the present invention. Further, in the present disclosure, like-named components of the implementations generally have similar features, and thus within a particular implementation each feature of each like-named component is not necessarily fully elaborated upon.

Approximating language, as used herein throughout the specification and claims, may be applied to modify any quantitative representation that could permissibly vary without resulting in a change in the basic function to which it is related. Accordingly, a value modified by a term or terms, such as “about,” “approximately,” and “substantially,” are not to be limited to the precise value specified. In at least some instances, the approximating language may correspond to the precision of an instrument for measuring the value. Here and throughout the specification and claims, range limitations may be combined and/or interchanged, such ranges are identified and include all the sub-ranges contained therein unless context or language indicates otherwise.

One skilled in the art will appreciate further features and advantages of the invention based on the above-described implementations. Accordingly, the present application is not to be limited by what has been particularly shown and described, except as indicated by the appended claims. All publications and references cited herein are expressly incorporated by reference in their entirety.

What is claimed is:

1. An ingredient container, comprising:
 - a container body having a hollow interior and an opening leading into the hollow interior;
 - a cap positioned over the opening in the container body and including an inlet port and an outlet port, wherein a cross-section of the cap extending substantially per-

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pendicular to a central axis of each of the inlet port and the outlet port has a shape that is a substantially circular triangle.

2. The ingredient container of claim 1, wherein the cap has an outer sidewall defining the shape of the cross-section and having first, second, and third walls.

3. The ingredient container of claim 2, wherein the first wall has a length that is greater than a length of each of the second and third walls.

4. The ingredient container of claim 2, the first wall is substantially planar, and the second and third walls are convex.

5. The ingredient container of claim 1, wherein the cap has a base wall with the inlet and outlet ports therein, and an outer sidewall surrounding the base wall, the outer sidewall having first and second shoulders projecting upward from the base wall.

6. The ingredient container of claim 2, wherein the cap includes a base wall having the inlet and outlet ports formed therein, and wherein the base wall includes a circular cavity formed therein at a mid-portion thereof.

7. The ingredient container of claim 6, wherein the inlet and outlet ports are positioned within the circular cavity.

8. An ingredient container, comprising:
 - a container body having a hollow interior and an opening leading into the hollow interior;
 - a lid coupled to the container body and covering the opening, the lid including an inlet leading from an exterior environment to the hollow interior and an outlet leading from the hollow interior to the exterior environment, the lid having a substantially triangular shape,

wherein the lid is configured to couple to a beverage dispenser having a complementary substantially triangular receiver.

9. The ingredient container of claim 8, wherein the lid includes a lid cover movable between an open position and a closed position, the lid cover being configured to close the inlet and the outlet when in the closed position.

10. The ingredient container of claim 8, wherein the lid includes a substantially circular inlet collar disposed around the inlet and a substantially circular outlet collar disposed around the outlet, the inlet collar and the outlet collar each protruding away from the lid.

11. The ingredient container of claim 10, wherein a height of the inlet collar is greater than a height of the outlet collar.

12. The ingredient container of claim 10, wherein the lid includes a lid cover movable between an open position and a closed position, the lid cover including a substantially circular inlet cover collar and a substantially circular outlet cover collar, the inlet collar being configured to encircle the inlet cover collar and the outlet collar being configured to encircle the outlet cover collar when the lid cover is in the closed position.

13. The ingredient container of claim 12, wherein the lid cover includes a central plug disposed within the outlet cover collar and configured to be received by the outlet.

14. The ingredient container of claim 8, wherein the lid has an outer sidewall defining the substantially triangular shape.

15. The ingredient container of claim 14, wherein the sidewall comprises first, second, and third walls, the first wall having a length that is greater than a length of each of the second and third walls.

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